
RAPTORGraph: Graph-Based Pathway Modeling for Causal Discovery in Single-Cell Perturbations

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Abstract

1 Experiments involving perturbation of single cells are central to understanding
2 cellular mechanisms and accelerating therapeutic discovery, however the space
3 of combinatorial perturbations is intractably large. Causal representation learn-
4 ing has shown great promise for predicting unseen combination of perturbations,
5 but existing methods often violate the theoretical requirements for identifiabil-
6 ity guarantees. We introduce RAPTORGraph, an end-to-end VAE framework
7 that addresses these issues through: i) a preconditioned *GraphPathway* encoder
8 that enforces intervention-guided mappings to causal meta-pathways, enabling
9 clean single latent-node interventions needed for causal identifiability, and ii)
10 optimal-transport alignment that aligns control and perturbed populations to stabi-
11 lize conditional generation. Empirical results on Perturb-seq datasets demonstrate
12 that RAPTORGraph improves the trade-off between reconstruction fidelity and dis-
13 tributional matching, and yields improved performance on predicting non-additive
14 combinatorial effects. Finally, we show that RAPTORGraph recovers biologically
15 meaningful latent programs and a causal graph over meta-pathways, providing an
16 interpretable bridge between generative quality and mechanistic insight.

17 1 Introduction

18 Understanding how cells respond to genetic perturbations, such as those induced by CRISPR
19 screens [Dixit et al., 2016], is fundamental to understand disease biology and advancing thera-
20 peutic discovery [Plenge et al., 2013, Adamson et al., 2016, Norman et al., 2019]. However, the
21 combinatorial explosion across $\sim 25,000$ genes makes exhaustive perturbation experiments infeasible
22 [Norman et al., 2019], motivating computational models that predict cellular responses to unseen
23 perturbations or combinations thereof.

24 Existing approaches to predict the transcriptional outcome following a perturbation face a fundamental
25 challenge: generative quality comes at the cost of causal interpretability. For example, recent methods
26 like GEARS [Roohani et al., 2024] or scGPT [Cui et al., 2024] achieve strong predictive performance
27 in cellular response to perturbations; however, they act as “black boxes” providing little mechanistic
28 insight into how perturbations affect cellular systems, failing to provide testable biological hypotheses.

29 Computational biology demands models that provide mechanistic insights into cellular function [Chen
30 et al., 2024], making causal representation learning (CRL) [Schölkopf et al., 2021] a promising ap-
31 proach for mapping (intervened) genes to biological processes. Recent CRL approaches such as
32 DiscrepancyVAE (dVAE) [Zhang et al., 2023] or SENA [de la Fuente et al., 2025] aim at recovering
33 the underlying causal factors aligned with the interventions, enabling counterfactual reasoning. How-
34 ever, these CRL approaches suffer from a fundamental architectural limitation: a dense interventional
35 encoder allows every gene, including the intervened ones, to influence many or all latent variables,
36 undermining the requirements for their identifiability guarantees. We term this phenomenon *interven-*

tion spillover, which renders identifying the unique latent target for a given intervention ill-posed. Consequently, the intervention mechanism cannot perform the clean, single-node manipulation required for valid causal inference [Brehmer et al., 2022, Lippe et al., 2022].

Further, the destructive nature of single-cell RNA sequencing (scRNA-seq) makes it impossible to observe the same cell before and after a genetic intervention, complicating the prediction of cell-specific responses. Existing methods address this issue by randomly pairing control and perturbed cells. Since current models are trained to minimize the error across all possible arbitrary pairings, they are incentivized to (suboptimally) learn the population-mean response. We refer to this phenomenon as *mean collapse*, as this regression to the mean erodes cellular diversity.

To address these critical issues, we present **Response Analysis of Perturbed TranscriptOmes using Interpretable Graph (RAPTORGraph)**, a novel framework that addresses causal identifiability challenges through a structured encoder design. Our first key innovation is the GraphPathway layer, a novel architecture that uses preconditioning to enforce sparse mappings from genes to learned interpretable causal factors, which we term causal meta-pathways. This design enables the model to perform the clean, single-node latent associations required for causal inference, while maintaining interpretability. This innovation is complemented by an optimal transport (OT) preprocessing step, which mitigates the distinct problem of mean collapse.

We benchmark the generative quality of RAPTORGraph on the Norman et al. [2019] Perturb-seq dataset and the Replogle et al. [2020] CRISPRi dataset, assess the accuracy of inferred genetic interactions under combinations of perturbations, and evaluate the biological interpretability of the inferred causal meta-pathways. We uncover biologically accurate programs in the K562 cell line, including the differentiation-induced cell cycle arrest driven by EGR1 and the stress-induced growth arrest mediated by JUN.

2 Background

2.1 Predicting transcriptional responses to unseen genetic perturbations

A key limitation in learning transcriptional responses to unseen genetic perturbations lies in the destructive nature of scRNA-seq, which prevents measuring the same cell before and after a perturbation. While paired measurements are infeasible, we do have access to independent single-cell observations from control and perturbed cells.

Recent works have demonstrated the potential of representation learning in leveraging perturbation data to predict transcriptional outcomes of novel combinations of perturbations. For example, Kamimoto et al. [2023] relied on inferring (linear) transcriptional relationships between genes in the form of a gene regulatory network to infer the effect of unseen perturbations. The Compositional Perturbation Autoencoder (CPA) [Lotfollahi et al., 2023] predicted combinations of perturbations by vector arithmetic in the latent space generated by a Variational Autoencoder (VAE). In parallel, GEARS [Roohani et al., 2024] leveraged Graph Neural Networks (GNNs) to incorporate prior knowledge of gene-gene relationships into the model architecture, thereby predicting unseen genetic interventions and their combinations. CellOT [Bunne et al., 2023] proposed to directly learn optimal transport maps between control and perturbed cell states, thus accounting for heterogeneous subpopulation structures in the data. Additional efforts, such as scGen [Lotfollahi et al., 2019], scGPT [Cui et al., 2024], have explored Transformer-based approaches for this task. However, the non-causal “black box” nature of the aforementioned models hampers their capability for uncovering the causal mechanisms driving transcriptional changes. This poses a significant limitation in, for example, cellular engineering, where understanding the *why* behind perturbation effects is as critical as predicting the *what*.

To address this challenge, dVAE [Zhang et al., 2023] learns causal factors without an explicit semantic meaning, resulting in causally valid but biologically uninterpretable latent representations. SENA [de la Fuente et al., 2025], on the other hand, explicitly models (a set of) biological processes as causal meta-pathways, promising both predictive accuracy and mechanistic interpretability. Further, these methods span a spectrum between fully exploratory and strongly confirmatory causal discovery approaches [Herrmann et al., 2024]. At one end, exploratory approaches such as dVAE learn all latent components directly from data. At the other, more confirmatory models such as SENA impose predefined biological pathways to improve interpretability. Although this latter strategy grounds the model in established biology, it can suffer from poor knowledge transfer: biological pathways are

highly cell-type- and cell-line-specific [Schneider et al., 2017, Hekselman and Yeger-Lotem, 2020], so mechanisms characterized in one cellular context may not generalize to another.

2.2 Modeling biological processes as latent causal variables

We model the underlying data-generating process using a structural causal model (SCM) [Pearl, 2009]. We assume that observed high-dimensional samples $\mathbf{x} \in \mathbb{R}^n$ (gene expression) are generated by a set of unobserved, low-dimensional *causal meta-pathways* $\mathbf{u} \in \mathbb{R}^d$, where $d \ll n$. Note that these causal meta-pathways may include several (related) known biological processes or pathways. The SCM defines the relationships between these variables via a set of structural assignments:

$$\mathbf{u}_i = f_i(\text{Pa}_{\mathcal{G}}(\mathbf{u}_i), \mathbf{z}_i), \quad \text{for } i = 1, \dots, d \quad (1)$$

where f_i is the structural function, $\text{Pa}_{\mathcal{G}}(\cdot)$ are the direct causal parents under graph $\mathcal{G}$, and $\mathbf{z}_i$ is an independent exogenous variable. The complete set of these equations induces a directed acyclic graph (DAG) $\mathcal{G}$, which implies that the joint probability distribution over $\mathbf{u}$ factorizes as:

$$p(\mathbf{u}) = \prod_{i=1}^d p(\mathbf{u}_i \mid \text{Pa}_{\mathcal{G}}(\mathbf{u}_i)) \quad (2)$$

However, these causal meta-pathways are not directly accessible. Instead, we only observe the high-dimensional data $\mathbf{x}$, which are generated via a true, unknown mixing function: $\mathbf{x} = g^*(\mathbf{u})$.

2.3 Identifiability of biological causal factors under genetic interventions

The central challenge of CRL is *identifiability*: learning an encoder function f_{θ} that can uniquely recover the true causal latents $\mathbf{u}$ from the observations $\mathbf{x}$. However, solely using observational data this goal is theoretically intractable, as multiple distinct causal structures can generate identical observational distributions $p(\mathbf{x})$ [Pearl, 2009, Khemakhem et al., 2020]. To overcome this key limitation, modern CRL relies on access to interventional data. The key insight is that interventions create unique statistical signatures that break the symmetries present in observational data (Fig. 1). Achieving identifiability depends on a set of key requirements [Schölkopf et al. [2021]]:

Requirement 1 (Acyclic Causal Structure). The causal structure is assumed to form a DAG, a foundational principle in causal discovery [Pearl, 2009].

Requirement 2 (Atomic Interventions). Causal structure recovery (i.e., identifiability guarantee) requires access to interventional data that targets each individual latent variable at least once. This requirement of atomic interventions is a key assumption in modern CRL [Zhang et al., 2023, von Kügelgen et al., 2023, de la Fuente et al., 2025].

Requirement 3 (Faithfulness). CRL also relies on the *faithfulness assumption*, which posits that all conditional independencies in the data are a consequence of the causal graph structure (d-separation) and not due to coincidental cancellations of parameters [Zhang et al. [2023]]

Under these requirements, it is possible to provably recover the true latent causal graph up to a well-defined equivalence class. Following the work of Zhang et al. [2023], their main theorem provides a formal guarantee for identifiability:

Theorem 2.1 (Full DAG Identifiability, Zhang et al. [2023]). *Assume a SCM with a latent DAG $\mathcal{G}$ and an invertible mixing function. If the set of interventions is atomic (targeting each latent variable at least once), and the assumptions of faithfulness hold, then the latent DAG $\mathcal{G}$ and the intervention assignments are identifiable up to permutation, scaling, and translation (the CD-equivalence class).*

The reliance on a **set of atomic interventions**, where each intervention targets a unique latent factor at least once, is the cornerstone of modern CRL-based prediction of genetic perturbations [Zhang et al., 2023, de la Fuente et al., 2025].

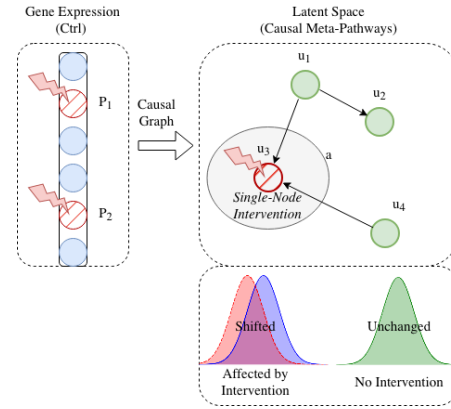


Figure 1: An intervention in gene expression $\mathbf{x}$ shifts the conditional distribution $p^I(\mathbf{u}_i \mid \text{Pa}_{\mathcal{G}}(\mathbf{u}_i))$ of a single meta-pathway $\mathbf{u}_i$ (causal latent factor), while leaving unrelated latent factors unaffected.

2.4 Intervention spillover in dense encoders: challenges for identifiability

While identifiability theory provides a strong theoretical foundation, a key challenge arises in recent CRL architectural choices: the *intervention spillover problem*. This phenomenon occurs when a single-variable intervention in the high-dimensional observation space $\mathbf{x}$ is transformed into a multi-variable distributional shift in the lower-dimensional latent space $\mathbf{u}$, which is in direct conflict with the atomic intervention requirement (Requirement 2 in Sec. 2.3). An *Ideal Causal Encoder*, f^* , must satisfy atomic intervention by ensuring that an intervention on a single gene generates a distributional effect on a sole latent factor, rather than a dense multi-variable distributional shift.

However, interventional encoders f_θ with dense architectures, like MLPs, possess a dense Jacobian matrix, which result in a dense output implicating multiple plausible latent targets. The model is thus forced to select a single “best” latent factor whose distribution to shift. This architectural choice has several critical implications for the reliability of the learned intervention targets. For a rigorous mathematical formulation of this ideal behavior, the formal assumptions, and the proof of incompatibility for dense architectures, we refer the reader to Sec. B.1.

Intervention spillover in current models for genetic perturbation prediction. This phenomenon is particularly concerning in genetic perturbation experiments, as interventions are applied to individual genes in the high-dimensional expression space, yet their effects propagate through the cell’s regulatory network and manifest as coordinated changes across multiple biological processes. **It is thus crucial to ensure, for each intervened gene, that the sole (causal) latent factor associated to it encompasses all relevant biological processes.** By enforcing the mapping to a unique latent through preconditioning encoder and a learned causal graph, our proposed architecture satisfies identifiability while still modeling complex gene-gene correlations.

3 RAPTORGraph: model architecture and learning

RAPTORGraph is an end-to-end framework designed to learn causal relationships from interventional single-cell data (Perturb-seq data). It consists of two novel modules: a preconditioned GraphPathway Encoder (Sec. 3.1) that maps genes to learned interpretable causal latent factors (the causal meta-pathways) and a DAGMA-based DAG module (Sec. 3.2) that infers the causal graph between the learnt causal meta-pathways. The proposed design directly addresses the Intervention Spillover Problem ensuring by construction, ensuring that gene interventions target a unique causal meta-pathway. We refer to Sec. C.1 for a detailed description of the model.

3.1 GraphPathway encoder with preconditioning

To address the Intervention Spillover Problem and enforce the theoretical requirement for single-causal-factor interventions, we introduce a novel encoder architecture, the GraphPathway Encoder, which features a deconfounding approach through preconditioning. This approach requires a specific ordering of the input data. Let k be the number of known perturbed genes in the experiment. Without loss of generality, we assume the input gene vector $\mathbf{x} \in \mathbb{R}^n$ is sorted such that the first k genes are those perturbed, followed by the remaining $n - k$ unperturbed genes.

Pre-conditioning module (Fig. 2-A). Given this sorted input, we define our model with d learned causal meta-pathways, where we require that the number of learned causal meta-pathways is at least the number of perturbed genes ($d \geq k$). We then structure the encoder’s primary weight matrix $\mathbf{W}$ as a block matrix that enforces a sparse, one-to-one mapping for perturbed genes while allowing dense interactions for the unperturbed components (see Sec. C.2 for the formal definition of $\mathbf{W}$).

This diagonal structure ensures each of the k perturbed genes directly influences only one of the first k learned causal meta-pathways (Fig. 2-A). The zero block in $\mathbf{W}$ is critical for preventing intervention spillover by explicitly blocking the influence of perturbed genes on the remaining learned causal meta-pathways. The other submatrices are learnable and dense, allowing the model flexibility to discover complex gene-gene relationships across all unperturbed genes and learned causal meta-pathways.

Crucially, this preconditioned structure allows the intervention mask $\mathbf{m}$, which specified which causal meta-pathway to shift for each intervention, to be predefined and fixed rather than learned, thereby satisfying the key structural requirements for identifiability.

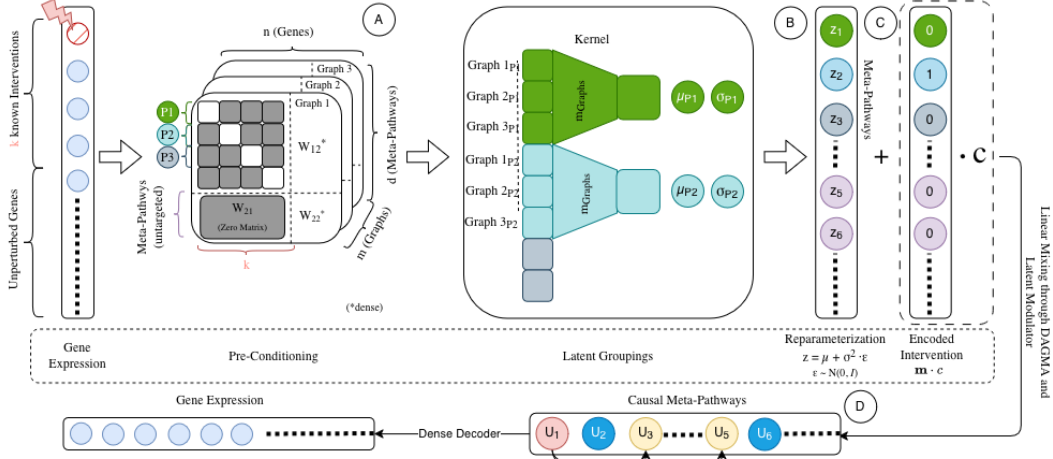


Figure 2: **The GraphPathway Encoder architecture of RAPTORGraph.** **A:** Gene expression of control cells (z_{obs}) is fed as input to the preconditioned linear layer, which enforces a sparse, one-to-one mapping for known perturbed genes to explicitly prevent intervention spillover. **B:** Complex dependencies are captured by learning multiple parallel graph representations, which are then aggregated to model non-linear interactions. **C:** A standard VAE reparameterization head generates the stochastic latent variables (z). **D:** A Latent Modulator takes the perturbation label as input and applies atomic perturbations (Δ) to the independent latents (z), shifting the distribution from the observed latent state (z_{obs}) to the interventional latent state (z_{int}). Both z_{obs} and z_{int} are then transformed by the DAGMA layer through the learned DAG via linear mixing to yield the final causal states (u), which are reconstructed into the predicted control ($\hat{x}_{\text{obs}}$) and perturbed ($\hat{x}_{\text{int}}$) gene expression profiles by a dense decoder.

188 **Latent grouping via an interaction block (Fig. 2-B).** To model the complex non-linear gene-gene
189 dynamics of biological systems, our architecture extends the preconditioning by learning multiple
190 distinct subgraph representations for each meta-pathway. An intermediate *Interaction Block* then
191 processes these subgraph activations, allowing the model to capture higher-order dependencies
192 by learning unique interaction patterns among the subgraphs within each learned meta-pathway
193 (Fig. 2-B).

194 **Meta-pathway inference via VAE (Fig. 2-C).** Next, an aggregation layer combines these processed
195 features to compute the VAE parameters (μ_z and σ_z) for the meta-pathway distributions (Fig. 2-
196 C). Thus, the final stochastic representations of the meta-pathways are then obtained using the
197 reparameterization trick $z = \mu_z + \sigma_z \odot \epsilon$; $\epsilon \sim \mathcal{N}(0, I)$.

198 **causal meta-pathway inference via DAGMA (Fig. 2-D, Sec. 3.2).** The model then represents the
199 downstream biological consequences of genetic interventions through the activities of causal meta-
200 pathways. Specifically, a DAGMA layer propagates the VAE-learned (independent) meta-pathway
201 activities through a learned directed acyclic graph, yielding the final causally dependent meta-pathway
202 activities.

203 As last step of the model, a non-linear dense decoder maps these causal states back to the high-
204 dimensional gene expression space to predict the post-perturbation (or unperturbed/control) gene
205 expression. For a comprehensive architectural description, we refer the reader to Sec. C.2.

206 3.2 Deep Causal Graph Learning with DAGMA

207 The GraphPathway Encoder with preconditioning fundamentally alters the causal structure discovery
208 task. By fixing the mapping to specific latent variables, it eliminates the need for permutation-based
209 discovery methods that assume a dense, entangled encoder. However, this fixed ordering makes
210 simple acyclicity constraints, such as enforcing an upper-triangular adjacency matrix, unsuitable.
211 These methods impose a strict and arbitrary causal hierarchy, as the influence of one latent variable
212 on another is entirely restricted by their ordering (e.g., latent i can only influence latent j if $i < j$),

which would conflict with the true biological relationships we aim to discover. For this, we adapt the DAGMA framework [Bello et al., 2022] for its use in a deep learning context.

DAGMA reformulates the combinatorial search for a DAG into a continuous optimization problem. This is achieved by minimizing a score function that measures data fit, subject to a differentiable constraint that is satisfied only if the graph is acyclic. The DAGMA acyclicity constraint, $h_s(\mathbf{A})$, is based on a log-determinant function:

$$h_s(\mathbf{A}) = -\log \det(s\mathbf{I} - \mathbf{A} \circ \mathbf{A}) + d \log s = 0 \quad (3)$$

where $\mathbf{A} \in \mathbb{R}^{d \times d}$ is the weighted adjacency matrix of the latent graph, $s > 0$ is a scaling scalar, and $\circ$ represents the Hadamard product. Note that the condition holds if and only if $\mathbf{A}$ represents a DAG. The optimization then solves a sequence of unconstrained problems that jointly minimize a data-fit score and the acyclicity penalty.

DAGMA implementation. We implement a modified DAGMA approach designed to learn the invariant, baseline causal structure of the biological system. To ensure stability and manage the trade-off between reconstruction fidelity and acyclicity, we introduce two critical regularization. First, we employ a path-following schedule where the score weight μ is gradually decreased (see Eq. (5)), allowing the model to prioritize feature learning early in training before strictly enforcing the graph constraint. Second, we isolate gradient flow by computing the DAGMA-associated loss $\mathcal{L}_{\text{DAGMA}}$ on detached latent representations. This ensures the acyclicity constraint optimizes the graph structure without distorting the encoder’s representation learning. A detailed description of the DAGMA layer, its objective, theoretical background, and implementation can be found in Sec. C.3.

3.3 Optimal Transport for Deterministic Interventional Learning

A fundamental challenge in single-cell interventional modeling is the lack of direct one-to-one mappings between control and perturbed cells, creating a counterfactual gap. Standard approaches often address this by randomly pairing control and interventional samples during training. However, this random pairing matches cells with poor counterfactuals, providing inconsistent gradient signals that force the model to minimize loss by predicting the average of the target distribution—a phenomenon we term *mean collapse* (see Fig. S5). To resolve this, we employ a Wasserstein-based OT framework to establish a deterministic, globally optimal mapping. By minimizing the total transport cost between the control and perturbed populations, OT pairs each cell with its most biologically similar counterpart. This preserves heterogeneous subpopulation structures and ensures stable, structure-preserving representations for causal discovery. Mathematical details are provided in Sec. E.

3.4 Training Objective

The model is trained end-to-end by minimizing a composite objective function, $\mathcal{L}_{\text{total}}$, which balances four specialized components to ensure robust representation learning and causal discovery:

$$\mathcal{L}_{\text{total}} = \alpha_{\text{rec}} \mathcal{L}_{\text{MSE}} + \beta_{\text{KL}} \mathcal{L}_{\text{KL}} + \gamma_{\text{MMD}} \mathcal{L}_{\text{MMD}} + \delta_{\text{DAGMA}} \mathcal{L}_{\text{DAGMA}} \quad (4)$$

where the scalar hyperparameters α_{rec} , β_{KL} , γ_{MMD} , and δ_{DAGMA} balance each term’s contribution. Specifically, we employ a reconstruction loss ($\mathcal{L}_{\text{MSE}}$) to maintain high-fidelity control unperturbed cell reconstruction; a KL divergence ($\mathcal{L}_{\text{KL}}$) to regularize the meta-pathway manifold; an interventional prediction loss ($\mathcal{L}_{\text{MMD}}$) to align distributions of predicted perturbed gene expressions with those of the experimental ground truth; and a causal graph loss ($\mathcal{L}_{\text{DAGMA}}$) to enforce structural acyclicity and data fit, defined as:

$$\mathcal{L}_{\text{DAGMA}} = \mu(S(\mathbf{A}; \mathbf{X}) + \lambda_1 \|\mathbf{A}\|_1) + h(\mathbf{A}) \quad (5)$$

where $S(\mathbf{A}; \mathbf{X})$ is a score (see Sec. C.3.2), $\|\mathbf{A}\|_1$ is the L1 norm of the adjacency matrix, and $h(\mathbf{A})$ is the acyclicity penalty. A comprehensive derivation and discussion of each term is provided in Sec. G.

4 Experiments

4.1 Evaluation Setup, Metrics and Datasets

Evaluation Metrics. To ensure a consistent and fair benchmark across all evaluated models, we use Maximum Mean Discrepancy (MMD) to assess the similarity between predicted and true gene

expression distributions. The MMD is calculated per type of perturbation (single or combinations thereof) and then averaged to provide an overall measure of distribution similarity. Alongside objective metric comparison, we use Precision@10 to evaluate non-additive interactions of combinatorial perturbations (see Sec. G).

Datasets. For evaluation we use two datasets: First, the Perturb-seq single-cell RNA-seq dataset from Norman et al. [2019] and as processed in the CPA study [Lotfollahi et al., 2023] and refer to it as the Norman-CPA dataset. It profiles 105 gene perturbations in K562 cells, including both single and double perturbations. In total, 284 conditions across $\sim 108,000$ cells were measured, of which 131 represent unique gene-gene combinations and the remaining 153 represent single perturbations. For each perturbation, associated control cells were also profiled. We utilize the entire set of single-perturbations during training and evaluate the subsequent methods on the double-perturbations.

Additionally, we used the CRISPRi single-cell RNA-seq dataset curated by Replogle et al. [2020] and as processed in the PerturBase study [Wei et al., 2024]. We refer to it as the Replogle2020 dataset. This dataset profiles 82 gene perturbations in K562 cells, including both single (44), double perturbations (37), and control. The 82 conditions were measured across $\sim 22,740$ cells.

Benchmarked models. To first establish the foundational accuracy of RAPTORGraph, we benchmarked its generative performance against four state-of-the-art models: SENA, scGPT, dVAE, and GEARS. We start by assessing two key aspects of prediction fidelity using the single-cell perturbation dataset from Norman et al. [2019]. We prioritized the Norman-CPA dataset for this comprehensive quantitative benchmarking because the implementations of several baseline models have auxiliary files and preprocessing steps that are tightly coupled to it, making adaptation to other datasets non-trivial. However, to further demonstrate generalizability, we provide an additional evaluation on the Replogle2020 dataset, restricted to a comparison between RAPTORGraph and scGPT.

4.2 Ablation Studies.

To validate our hyperparameter selection, we conducted a comprehensive ablation study on the VAE regularization parameter (β) and the DAGMA score weights (μ, λ_1). Our analysis reveals a critical trade-off between distribution matching and the preservation of biological heterogeneity. Detailed results of the full hyperparameter sweep and the rationale for the selected configuration are provided in Sec. H.6.

4.3 Reconstruction capabilities.

First, we measured the Mean Squared Error (MSE) between the ground truth and predicted profiles of control cells to evaluate how well each model reconstructs the basal, unperturbed cellular state. Second, we calculated the MMD between the distributions of ground truth and predicted profiles for double-gene perturbations to assess the model’s ability to capture the global transcriptional shift following a complex intervention. A detailed description of these metrics is provided in Sec. G.

Table 1: Results using the Norman-CPA dataset are averaged over 3 runs. Bold: best results; Italics: second best. Values are mean $\pm$ variance. *We excluded the GEARS model because it behaves in a fundamentally different way that is incompatible with our evaluation framework, making a fair and direct comparison impossible (see Sec. G.8).

Metric	SENA	scGPT	dVAE	GEARS	RAPTORGraph (ours)
MMD	0.522 $\pm$ 0.003	0.176 $\pm$ 0.001	0.076 $\pm$ 0.000	—*	<i>0.112 $\pm$ 0.001</i>
MSE	0.043 $\pm$ 0.000	0.066 $\pm$ 0.000	0.062 $\pm$ 0.000	—*	<i>0.045 $\pm$ 0.000</i>

The results in Table 1 highlight a crucial trade-off between reconstruction fidelity (MSE) and distributional accuracy (MMD). RAPTORGraph achieves a strong overall balance, demonstrating highly competitive performance across both reconstruction and generative capabilities.

The additional benchmark using the Replogle2020 dataset (shown in Table 2 and restricted to a direct comparison between RAPTORGraph and scGPT) further demonstrates our model’s robust performance and superior reconstruction fidelity on independent biological data.

4.4 Prediction of non-additive genetic perturbations.

This complex and biologically significant benchmark assesses a model’s ability to predict complex interaction effects between single-gene perturbations that are beyond simple linear combinations (Table 3). Specifically, we compared each model based on their ability to predict five distinct interaction types, using Precision@10 to quantify how effectively each model could identify true interactions within its top 10 predictions, a direct measure of its utility for experimental discovery (see Sec. G.5). RAPTORGraph is among the top performers in almost all metrics, yielding the best overall performance across all methods.

Table 2: Results using the Replogle2020 dataset are averaged over 3 runs. Values are mean $\pm$ variance. Bold: best results.

Metric	scGPT	RAPTORGraph (ours)
MMD	0.138 $\pm$ 0.000	0.137 $\pm$ 0.000
MSE	0.069 $\pm$ 0.000	0.054 $\pm$ 0.000

Table 3: Comparison of non-additive genetic interaction prediction. Precision@10 scores for five baseline models across distinct genetic interaction subtypes on the Norman et al. dataset. Higher is better. Bold: best results; Italics: second best. Values are mean $\pm$ variance over 3 runs.

Metric	dVAE	SENA	GEARS	scGPT	RAPTORGraph (ours)
Synergy	0.000 $\pm$ 0.000	0.233 $\pm$ 0.153	<i>0.333 $\pm$ 0.115</i>	0.100 $\pm$ 0.000	0.367 $\pm$ 0.058
Suppression	0.000 $\pm$ 0.000	0.167 $\pm$ 0.115	<i>0.333 $\pm$ 0.115</i>	0.400 $\pm$ 0.000	0.267 $\pm$ 0.153
Neomorphism	<i>0.333 $\pm$ 0.153</i>	<i>0.333 $\pm$ 0.115</i>	0.133 $\pm$ 0.058	0.100 $\pm$ 0.000	0.433 $\pm$ 0.058
Redundancy	0.933 $\pm$ 0.115	0.467 $\pm$ 0.058	0.567 $\pm$ 0.115	0.100 $\pm$ 0.000	<i>0.800 $\pm$ 0.100</i>
Epistasis	<i>0.567 $\pm$ 0.153</i>	0.800 $\pm$ 0.100	0.533 $\pm$ 0.115	0.400 $\pm$ 0.000	0.433 $\pm$ 0.115

OT ablation study. We further investigated the contribution of OT to this performance. Our ablation study reveal that OT preprocessing is a key driver of the model’s ability to predict complex non-additive interactions, particularly boosting redundancy and epistasis. Importantly, OT-based pairing significantly outperforms a random pairing baseline. Note that this benefit comes with negligible computational overhead ($\approx 2.2\%$ of total training time). Detailed analyses of these empirical benefits and the computational cost are provided in Sec. H.5 and Sec. H.3, respectively.

4.5 Reverse Perturbation Analysis

While predicting genetic interactions demonstrates a model’s forward predictive power, a more profound test is its ability to reverse-engineer the genetic cause of an observed phenotype. To this end, we challenged a model to identify the specific genetic perturbation responsible for a given cellular gene expression profile. The analysis was performed on a controlled combinatorial space of 20 transcription factors from the Norman et al. [2019] dataset. For each model, we generated a database of predicted expression profiles for all possible double-gene perturbations within this subset. We then used the ground truth mean expression profiles from experimentally observed double perturbations as queries, ranking the database entries by similarity to identify the most likely causal perturbation.

We first evaluated the models on the highly stringent task of identifying the exact causal gene combination, quantified by the Correct (Exact Match) Hit Rate @ K. As shown in Fig. S6, all models achieve modest hit rates, underscoring the profound difficulty of this reverse-engineering challenge.

However, a more practical measure of a model’s utility to guide experimental genetic perturbation designs is its ability to retrieve perturbations containing at least one of the correct genes. We thus evaluated each model’s capacity to identify biologically influential genes using the Relevant (Gene Overlap) Hit Rate @ K (see Fig. 3 and Table 4).

RAPTORGraph demonstrates exceptional performance, consistently ranking among the top models. This highlights that while pinpointing exact higher-order interactions is an ongoing challenge, RAPTORGraph is highly effective at identifying the key genetic players driving a cellular response, making it a highly valuable tool for hypothesis generation and experimental design.

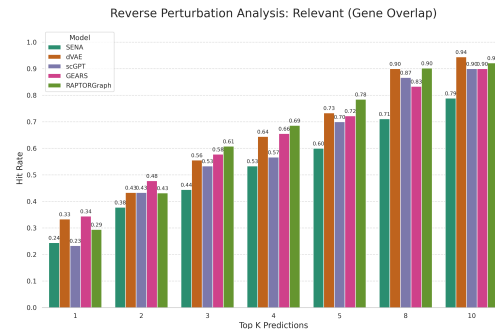


Figure 3: Hit Rate @ K (relevant) for the reverse perturbation task. The metric measures the frac-

Table 4: Hit Rate @ K (relevant) for the reverse perturbation task. The metric measures the fraction of queries where at least one correct gene was identified in the top K predictions. Higher values are better. Values are mean $\pm$ variance over 3 runs.

k	dVAE	SENA	GEARS	scGPT	RAPTORGraph (ours)
1	0.333 ± 0.115	0.244 ± 0.019	0.344 ± 0.051	0.233 ± 0.000	0.294 ± 0.059
3	0.556 ± 0.051	0.444 ± 0.069	0.578 ± 0.051	0.533 ± 0.000	0.608 ± 0.122
5	0.733 ± 0.088	0.600 ± 0.058	0.722 ± 0.077	0.700 ± 0.000	0.784 ± 0.090

5 RAPTORGraph learns biologically meaningful causal meta-pathways.

To confirm that RAPTORGraph learns biologically meaningful causal structures rather than arbitrary representations, we examined the functional identity of the causal meta-pathways using Gene Set Enrichment Analysis (GSEA). For each causal meta-pathway, we identify the N single-gene interventions that induce the greatest change in that meta-pathway. Then, for every such ranked gene list, we perform a GSEA to identify significantly enriched biological processes (also known as pathways) within this set of N genes, serving as transcriptional signatures for the causal meta-pathway. It is important to differentiate between the GSEA-derived pathways (which are well-described biological processes from known databases) and the causal meta-pathways (which are learned causal latent factors that are associated with single interventions). Thus, our goal here is to assess whether the learned causal meta-pathways encompass relevant and related biological processes. For this evaluation, we use the ‘‘Hallmark’’ gene sets from the Human Molecular Signatures Database (MSigDB) database [Liberzon et al., 2015].

The GSEA analysis reveals that RAPTORGraph captures distinct and biologically accurate programs. Figure S7 visualizes the Normalized Enrichment Scores (NES) for the top enriched pathways across the learned factors. We highlight distinct biological programs captured by the model, verified against the known functions of their corresponding gene perturbations:

- Differentiation-Induced Arrest:** The causal meta-pathway linked to *EGR1* displays massive negative enrichment for ‘‘E2F Targets’’ and ‘‘G2-M Checkpoint’’. *EGR1* is a known tumor suppressor in K562 leukemia cells that drives differentiation along the megakaryocytic lineage, a process that necessitates a complete exit from the cell cycle [Ma et al., 2019]. RAPTORGraph correctly identifies this potent anti-proliferative program.
- Stress-Induced Growth Arrest:** The causal meta-pathway linked to the AP-1 subunit *JUN* shows strong negative enrichment for ‘‘E2F Targets’’ and ‘‘G2-M Checkpoint’’. While often associated with inflammation, c-Jun is a central stress-response mediator; in the context of K562 cells, its activation drives a termination of cell division required for stress adaptation or differentiation [Shaulian and Karin, 2002, Shaulian et al., 2000].
- p53-Like Tumor Suppression:** The causal meta-pathway associated with *TP73* displays significant negative enrichment for ‘‘G2-M Checkpoint’’. *TP73* is a functional homolog of *TP53* (p53); its activation triggers downstream checkpoint signaling that arrests the cell cycle to prevent genomic instability [Kaghad et al., 1997].

These findings confirm that the nodes in our learned DAG represent specific, identifiable biological mechanisms. Consequently, the edges learned by the DAGMA module can be interpreted as causal regulatory links between these verified biological processes. A detailed description of the methodology is provided in Sec. G.7. Detailed GSEA results and the associated enrichment heatmap are provided in Sec. H.2.

6 Conclusion

Developing interpretable *in silico* models is crucial for therapeutic discovery, yet current approaches face two key challenges: intervention spillover, where dense encoders create entangled latent signals that confound causal discovery, and mean collapse, where the absence of one-to-one cell pairings erodes cellular diversity. We introduce RAPTORGraph, a framework that directly resolves these issues. To counteract intervention spillover, it combines a preconditioned GraphPathway layer, enforcing the sparse mappings required for clean single-node interventions, with a DAGMA layer for robust DAG discovery. This architectural design eliminates the artifactual entanglement of input genes with latent pathways, while the subsequent DAG module explicitly learns the legitimate biological causal relationships and correlations between the resulting sparse pathway activities. To prevent mean collapse, RAPTORGraph globally pairs control and perturbed populations through OT. RAPTORGraph not only achieves a superior balance between reconstruction fidelity and distributional accuracy compared to state of the arts, but also demonstrates a superior capability to predict complex, non-additive genetic interactions, and can reverse-engineer genetic drivers to generate testable hypotheses. RAPTORGraph bridges predictive power and mechanistic insight, enabling not only the prediction of a perturbation's effect but also understanding the reasoning behind it. Future work will focus on deploying RAPTORGraph for the hypothesis-driven discovery of novel drug targets and exploring its utility in uncovering cell-type specific pathway mechanisms.

Broader Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning by improving the interpretability and accuracy of causal discovery in biological systems. By enabling more precise *in silico* predictions of genetic perturbations, our framework potentially accelerates therapeutic discovery and reduces the need for exhaustive experimental screening. While the primary focus is on basic research in cellular biology, we recognize that improved causal modeling of biological data could eventually inform clinical decision-making; thus, we emphasize the importance of independent experimental validation before applying these findings in medical contexts. There are no direct negative societal consequences of this work anticipated at this stage.

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A Nomenclature and Mathematical Notation

In this section, we provide a comprehensive list of the mathematical notations used throughout our work, organized by category. This serves as a centralized reference to ensure clarity and consistency.

A.1 General Variables and Data Structures

We begin by defining the fundamental variables and data structures used to represent observed and latent spaces.

Symbol	Description
$\mathbb{R}$	The set of real numbers.
$\mathbf{x}$	A vector of observed variables (e.g., gene expression).
$\mathbf{z}$	A vector of general latent variables in the learned representation space.
$\mathbf{e}$	A vector of general, unspecified noise terms.
$\mathbf{X}$	A matrix representing a set of observed data samples.
$\mathbf{I}$	The identity matrix.

A.2 Causal Models and Graphs

Next, we define the components of the Structural Causal Model (SCM) that describes the underlying data-generating process.

Symbol	Description
$\mathcal{G}$	A DAG representing the causal structure.
$\mathbf{A}$	The weighted adjacency matrix of the causal graph $\mathcal{G}$.
$\mathbf{B}$	A general adjacency matrix.
$\mathbf{W}$	A general weight matrix.
$\text{Pa}_{\mathcal{G}}(\cdot)$	A function that returns the parent nodes of a given node in $\mathcal{G}$.
$\bar{\mathcal{G}}$	The transitive closure of the graph $\mathcal{G}$.
$\mathbf{u}$	The true, unobserved latent causal variables in the SCM.
u	A single latent causal variable.
$\mathbf{z}$	A vector of exogenous noise terms for the SCM.
f_i	The structural causal function for a single variable u_i .
$\text{Tr}(\cdot)$	The trace function for a matrix.

A.3 Model Functions and Representations

We denote the core functions of our model, which learn to map between the observed and latent spaces, as follows.

Symbol	Description
g^*	The true, unknown data-generating (mixing) function.
g	The learned decoder network.
f^*	The ideal, oracle encoder function.
f_{θ}	The practical, parameterized encoder network.
$\hat{\mathbf{z}}$	The learned latent representation, output of the encoder ($f_{\theta}(\mathbf{x})$).
$\hat{\mathbf{x}}$	A single data sample generated by the model’s decoder.
$\hat{\mathbf{X}}$	A matrix representing a batch of data samples generated by the model’s decoder.
a	The composed function $f_{\theta} \circ g^*$.
$\tilde{\mathbf{z}}$	The true latent representation produced by the f^* .
$\Delta\tilde{\mathbf{z}}$	The ideal, single-node change in the latent space.
$\Delta\mathbf{z}$	The approximated (dense) change in the latent space.
$\mathcal{N}$	The normal (Gaussian) distribution.

545 A.4 Intervention-Specific Variables

546 Here, we list the variables specifically related to the modeling of interventions.

Symbol	Description
q_{int}	The intervention encoder network.
$\mathbf{i}$	The one-hot vector specifying the target of an intervention.
$\mathbf{c}$	A general condition vector input to an intervention network.
$\mathbf{m}$	The mask vector specifying an intervention's target(s).
c	The scalar strength of a latent intervention.
$\mathbf{z}_{\text{obs}}$	Latents from an observational (control) sample.
$\mathbf{z}_{\text{int}}$	Latents from an interventional sample.
$P_Z^{(i)}$	The interventional distribution over latent variables.
I_k	The set of intervened variables in environment k .
$\mathbf{t}_i$	The targets of an intervention for a variable i .
$\mathbf{u}$	An auxiliary variable providing conditioning information (e.g., environment index).

547 A.5 Identifiability and Transformations

548 We define symbols used in the context of identifiability proofs and the resulting transformation
549 classes.

Symbol	Description
$\sim_T$	Equivalence class for a transformation T.
$\sim_C$	Equivalence class for a transformation C.
$\mathbf{A}$	The matrix component of an affine transformation.
$\mathbf{c}$	The vector component of an affine transformation.
e	A scalar factor in a scaling transformation.
b	An offset in a shift transformation.

550 A.6 Probability, Losses, and Functions

551 We define the probability functions, loss functions, and mathematical operators used to train our
552 models and enforce constraints.

Symbol	Description
$P(\cdot)$	PMF (for discrete variables).
$p(\cdot)$	PDF (for continuous variables).
$F(\cdot)$	CDF.
p_{data}	The true, underlying data-generating distribution (PDF).
p_{model}	The learned distribution of the model (PDF).
$\mathbb{E}[\cdot]$	The expectation function.
$\mathcal{L}_{\text{DAGMA}}$	Loss function related to the causal graph structure.
$\mathcal{L}_{\text{MMD}}$	Loss function based on the MMD.
h	A function used to enforce graph acyclicity.
S	A score function for graph discovery.
$\text{MMD}(\cdot, \cdot)$	The MMD function, which computes the distance between two distributions.
$\text{MMD}(\cdot, \cdot)$	The MMD function, which computes the distance between two distributions.
$k(\cdot, \cdot)$	A kernel function used in MMD computation.

553 A.7 Hyperparameters

554 We list the key hyperparameters that control our model's training and behavior.

Symbol	Description
λ_{score}	Regularization hyperparameter for a graph score.
σ_{MMD}	The bandwidth hyperparameter for the Gaussian RBF kernel.
β	The weights for the convex combination in multi-kernel MMD.
L	The number of kernels used in the MMD calculation.
m	The number of graphs.
i	The number of samples in a batch of observed data.
j	The number of samples in a batch of predicted data.

A.8 Downstream Analysis Metrics

We define variables used in the downstream evaluation tasks, such as the genetic interaction analysis.

Symbol	Description
Δ	The perturbation effect vector (mean of perturbed minus mean of control).
$\Delta(A + B)_{\text{naive}}$	The naive additive effect vector, defined as $\Delta A + \Delta B$.
$\ \cdot\ _2$	The L2 norm of a vector.
$\cos(\cdot, \cdot)$	The cosine similarity between two vectors.
$\bar{\mathbf{x}}$	The mean expression profile of a cell population.
N	The number of cells for a given perturbation condition.
f_M	The prediction function for a given model M .
Synergy Ratio	The score used to measure synergy and suppression.
Neomorphism Score	The score used to measure neomorphism.
Redundancy Score	The score used to measure redundancy.
Epistasis Score	The score used to measure epistasis.

A.9 Dimensionality and Spaces

Finally, we list symbols for dimensionality and the properties of the data spaces.

Symbol	Description
n	The dimensionality of the observed data space.
d	The dimensionality of the latent space.
d	The number of learned pathways, equivalent to d .
p	The degree of a polynomial mixing function.
$\mathcal{Z}$	The support of the latent distribution.
$\mathcal{X}$	The support of the observed distribution.
n	The number of genes in the expression profile, equivalent to n .

A.10 Acronyms and Abbreviations

B Mathematical Proofs and Theoretical Background

This appendix provides a detailed theoretical examination of the core challenges in causal representation learning that motivate our proposed architecture. We begin by formally defining and proving the incompatibility of dense encoders via the Intervention Spillover Problem, and then discuss the role of prior knowledge in differentiating causal modeling strategies.

B.1 The Intervention Spillover Problem

This section provides the formal theoretical background for the *Intervention Spillover Problem*. We formally define this problem, arguing that it is a direct consequence of the inherent properties of dense encoder architectures commonly used in causal representation learning (CRL). We then prove the fundamental incompatibility between these architectures and the requirements for causal identifiability.

Definition B.1 (The Intervention Spillover Problem). The *Intervention Spillover Problem* is the phenomenon where a targeted single-variable intervention in a high-dimensional observation space is transformed into a dense, multi-node signal in the lower-dimensional latent space. This issue arises as a direct consequence of using a practical, dense encoder architecture, which inevitably corrupts the intervention’s sparsity and creates a fundamental incompatibility with CRL methods that require sparse, single-node latent interventions for identifiability.

B.1.1 Mathematical Preliminaries and Setup

While the problem described is general, this analysis formalizes it within the context of a *linear SCM* for clarity. Let the gene expression space be $\mathbb{R}^n$ and the latent space be $\mathbb{R}^d$, with $d \ll n$. Let $f_\theta : \mathbb{R}^n \rightarrow \mathbb{R}^d$ be the practical, differentiable encoder network we train. It maps an observation vector $\mathbf{x}$ to an *approximated* latent vector $\mathbf{z} = f_\theta(\mathbf{x})$. The local behavior of this encoder is described by its Jacobian matrix $\mathbf{J}_\theta(\mathbf{x}) \in \mathbb{R}^{d \times n}$.

We model the latent space with a linear SCM, where causal relationships are defined by an adjacency matrix $\mathbf{A}$. An intervention on a single gene creates a post-intervention vector $\mathbf{x}_{\text{int}}$ from a corresponding control vector $\mathbf{x}_{\text{obs}}$. This experimental paradigm is characteristic of modern pooled CRISPR screens (e.g., Perturb-seq), where the genetic target of each intervention is known by design.

B.1.2 Formal Assumptions

Assumption B.2 (Ideal Causal Encoder and Single-Node Intervention Target). An *Ideal Causal Encoder*, f^* , is an unknown oracle function that perfectly disentangles the effect of any intervention. The key property is that the *true* latent change, $\Delta\tilde{\mathbf{z}} = f^*(\mathbf{x}_{\text{int}}) - f^*(\mathbf{x}_{\text{obs}})$, corresponds to a single-node perturbation. This ideal latent change is defined as $\Delta\tilde{\mathbf{z}} = \mathbf{m} \cdot c$, where c is the latent intervention strength and the target mask $\mathbf{m} \in \{0, 1\}^d$ is *one-hot*:

$$\sum_{i=1}^d m_i = 1 \tag{S1}$$

In a *closed-world experimental setting*, where the targeted gene is known *a priori*, the behavior of the f^* must fulfill the single-node intervention requirement. The following lemma formalizes this ideal mapping.

Lemma B.3 (Equivalence in Ideal Latent Space). *For an Ideal Causal Encoder f^* operating in a closed-world setting, the latent representation of the post-intervention sample is equivalent to the latent representation of the control sample plus the ideal single-node latent perturbation. Formally:*

$$f^*(\mathbf{x}_{\text{int}}) = f^*(\mathbf{x}_{\text{obs}}) + \mathbf{m} \cdot c \tag{S2}$$

Proof. The proof follows directly from the definition of the ideal latent change in theorem B.2. We start with the identity $\Delta\tilde{\mathbf{z}} = f^*(\mathbf{x}_{\text{int}}) - f^*(\mathbf{x}_{\text{obs}})$. Substituting the single-node perturbation gives $f^*(\mathbf{x}_{\text{int}}) - f^*(\mathbf{x}_{\text{obs}}) = \mathbf{m} \cdot c$. Rearranging the terms yields the desired result. $\square$

602 **Assumption B.4** (Practical Model Approximation and Dense Jacobian). Our *Practical Model* uses
 603 an encoder $f_\theta(\mathbf{x})$ and a perturbation inference network q_{int} . The underlying encoder network has a
 604 *dense Jacobian matrix* $\mathbf{J}_\theta(\mathbf{x})$.

605 **Remark B.5** (Architectural Bias vs. Learned Function). It is important to distinguish between the
 606 architecture of the encoder and the function it learns. While it is theoretically possible for a densely-
 607 connected network to learn an effectively sparse Jacobian, the architecture itself possesses a strong
 608 inductive bias towards dense mappings. Standard training objectives provide no direct incentive for
 609 learning a sparse function.

610 B.1.3 Main Result: The Incompatibility

611 **Lemma B.6** (Incompatibility of the Practical Model). *Given the practical model's reliance on a*
 612 *dense encoder network (theorem B.4), there is no guarantee that an intervention on a single gene*
 613 *will produce a single-node latent intervention target vector $\mathbf{m}$ as required by the ideal model in*
 614 *theorem B.2.*

615 *Proof.* The proof rests on the direct conflict between the output of the practical encoder and the
 616 requirements of the ideal model in a closed-world setting. In this setting, we have access to both
 617 the control sample $\mathbf{x}_{\text{obs}}$ and the post-intervention sample $\mathbf{x}_{\text{int}}$. Our trained practical encoder, f_θ , is a
 618 known deterministic function (e.g., a trained MLP).

619 When we process these known samples, the encoder produces an approximated latent change:

$$\Delta \mathbf{z} = f_\theta(\mathbf{x}_{\text{int}}) - f_\theta(\mathbf{x}_{\text{obs}}) \quad (\text{S3})$$

620 Due to the dense Jacobian of f_θ (theorem B.4), each component of the latent vector $\mathbf{z}$ is a function of
 621 all input components in $\mathbf{x}$. Consequently, a change in a single gene will result in a change across
 622 nearly all latent dimensions. The resulting vector $\Delta \mathbf{z}$ is therefore dense.

623 The objective for a conditional network q_{int} is to learn to predict this latent change. For the model to
 624 perfectly capture the intervention effect, the learned perturbation $(\mathbf{m} \cdot \mathbf{c})$ must match the observed
 625 latent change. This implies minimizing the loss:

$$\mathcal{L} = \|\Delta \mathbf{z} - (\mathbf{m} \cdot \mathbf{c})\|^2 \quad (\text{S4})$$

626 To achieve a loss of zero, the learned perturbation $(\mathbf{m} \cdot \mathbf{c})$ must be equal to the dense vector $\Delta \mathbf{z}$. This
 627 forces the learned target mask $\mathbf{m}$ to be dense, which directly contradicts the one-hot requirement
 628 for a single-node intervention as defined in Eq. (S1). Therefore, the practical model's architecture is
 629 fundamentally misaligned with the goal of identifying single-node latent interventions. $\square$

630 B.1.4 Implications for Causal Discovery in Dense Encoders

631 This mathematical reality creates a cascade of problems for causal discovery models that rely on a
 632 learned intervention network to identify the latent target of a perturbation. The core of the issue is the
 633 need to make a discrete, single-target selection. This is typically addressed using a temperature-scaled
 634 softmax function, a mechanism that mimics the Gumbel-Softmax trick for differentiable categorical
 635 sampling [Jang et al., 2016]. Given the output logits, $\mathbf{o}$, from an intervention network, the probability
 636 for each target is calculated as:

$$\pi_i = \frac{\exp(\mathbf{o}_i/\tau)}{\sum_{k=1}^d \exp(\mathbf{o}_k/\tau)} \quad (\text{S5})$$

637 where $\tau > 0$ is a temperature hyperparameter that controls the sharpness of the output distribution.

638 As the temperature $\tau \rightarrow 0$, the softmax output approaches a discrete, one-hot vector (a proxy for
 639 a Dirac delta function), representing a single, "hard" choice. However, this creates a fundamental
 640 contradiction when applied to the output of a dense encoder. The encoder produces a dense latent
 641 change vector, $\Delta \mathbf{z}$, where the evidence suggests a multi-node perturbation. To satisfy the model's
 642 objective of selecting a single target, a low temperature τ must be used to force the softmax into
 643 a "winner-take-all" state. This forces the model to ignore the dense evidence and declare a single
 644 winner, which can lead to it learning spurious, causally unfaithful mappings.

645 The temperature-scaled softmax is therefore caught in an architectural dissonance: it is a mechanism
 646 forced to make a discrete choice that fundamentally misrepresents the dense, continuous signal it
 647 receives from the encoder.

Furthermore, in the closed-world setting of a Perturb-seq experiment, the mapping from a perturbed gene to its corresponding latent variable is known beforehand. Therefore, the task of “exploratory perturbation” (i.e., discovering which latent variable was the target) is not actually necessary. The true challenge is to learn the *scale* or *power* of the intervention’s effect on the causal representation, not its target, a task for which these architectures are ill-suited.

B.2 The Role of Prior Knowledge in Causal Modeling

The theoretical gaps in existing models can be understood by positioning them on a spectrum between two complementary modes of scientific inquiry: exploratory and confirmatory research [Herrmann et al., 2024]. Exploratory research is concerned with hypothesis generation; it involves analyzing data to uncover novel patterns, structures, and relationships without a strong pre-existing theory. In contrast, confirmatory research is concerned with hypothesis testing; it begins with a specific, pre-formulated hypothesis and uses data to determine if that hypothesis is supported or refuted. This distinction helps clarify the intended purpose and inherent trade-offs of different causal models. For instance, **DiscrepancyVAE (dVAE)** is primarily exploratory, aiming to discover latent variables and their causal graph from scratch [Zhang et al., 2023]. In contrast, **SENA** takes a more targeted approach, acting as a confirmatory model for pathways while exploring perturbation effects; it **confirms** latent variables as known biological pathways and **explores** which pathway is targeted by a perturbation [de la Fuente et al., 2025].

The approach by de la Fuente et al. [2025] highlights the close relationship between SCM and CRL. By assuming its causal variables (the biological pathways) are known *a priori*, it operates like a traditional SCM. However, because it must still *learn* a representation of how to map raw gene expression data onto these variables and then discover the causal graph between them, it firmly resides within the CRL paradigm. It represents a specific, constrained form of CRL where prior knowledge heavily guides the representation learning process. This confirmatory design choice is motivated by the goal of enhancing the biological interpretability of the learned causal model.

However, this reliance on predefined pathways introduces significant limitations, primarily stemming from the challenge of knowledge transfer. Biological pathways exhibit strong cell-type and cell-line specificity, meaning that mechanisms characterized in one cellular context often do not generalize to another [Schneider et al., 2017, Gamazon et al., 2018, Hekselman and Yeger-Lotem, 2020]. This context dependence poses a fundamental barrier to transferring pathway insights across different *in vitro* models and ultimately to patient tissues [Walter, 2019]. Consequently, the findings from confirmatory models are difficult to apply to less-researched cell types, which limits the scalability and broader applicability of these approaches. Furthermore, this confirmatory strategy may not be suitable for smaller-scale experiments that only incorporate a limited set of genes, making it difficult to map observations to comprehensive pathway definitions.

C Model Architecture

This section provides a comprehensive overview of the core architectural components of our model, detailing both the novel GraphPathway Encoder and the deep causal graph learning framework.

C.1 Overview of RAPTORGraph framework

The architecture of the **R**esponse **A**nalysis of **P**erturbed **T**ranscript**O**mes using **I**nte**R**pretable **G**raph (RAPTORGraph) framework, illustrated in Fig. S4, is designed as an end-to-end system for learning causal relationships from interventional single-cell data. The framework is composed of three primary modules: a preconditioned encoder for sparse representation learning, a DAG discovery layer for identifying causal structure, and a decoder for reconstructing cellular states.

The input to the model is the high-dimensional gene expression data. The first layer of the encoder is our novel preconditioned GraphPathway Encoder, which addresses the Intervention Spillover Problem. By leveraging prior biological knowledge, it enforces a sparse mapping from the thousands of observed genes ($\mathbb{R}^n$) to a lower-dimensional set of learned meta-pathways ($\mathbb{R}^d$). This ensures that a single-gene intervention activates only a sparse, well-defined set of learned meta-pathways in the latent space, fulfilling the requirements for valid downstream causal inference.

The resulting latent representations, $\mathbf{z}$, are then passed to the DAGMA-based DAG discovery module. This layer operates directly on the latent variables to learn the causal relationships *between* the learned meta-pathways. Its primary output is a directed acyclic graph ($\mathcal{G}$), represented by its adjacency matrix, which models the flow of biological influence among the learned meta-pathways.

Finally, the decoder network reconstructs the original gene expression profile from the latent representation. This ensures that the latent variables, in addition to being causally structured, also retain sufficient information to accurately capture the full cellular state. The model is trained end-to-end by optimizing a composite loss function that balances these different objectives, as detailed in the figure caption.

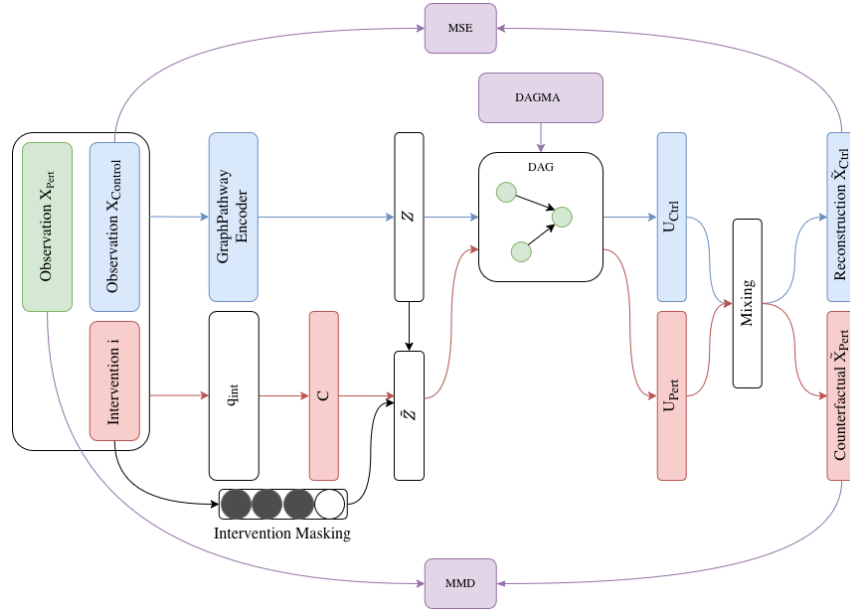


Figure S4: Detailed schematic of the RAPTORGraph framework. The preconditioned GraphPathway Encoder enforces a sparse mapping from the high-dimensional gene expression space ($\mathbb{R}^n$) to the learned meta-pathways ($\mathbb{R}^d$). The DAGMA-based DAG operates on the latent representations to learn a directed acyclic graph ($\mathcal{G}$) representing the causal relationships between the learned meta-pathways. The model is trained end-to-end using a composite loss function comprised of several terms: a reconstruction loss (Mean Squared Error (MSE)), a distributional loss (MMD) on interventional predictions, a KL Divergence on the Variational Autoencoder (VAE) latent variables, and an acyclicity constraint from the DAGMA layer.

707 C.2 GraphPathway Encoder Architecture

708 The GraphPathway Encoder is a custom neural network layer designed to solve the Intervention
709 Spillover Problem through a principled architectural design. It combines a preconditioned weight
710 matrix to deconfound known perturbation effects with a multi-subgraph structure to capture complex,
711 non-linear gene interactions.

712 C.2.1 Deconfounding through Preconditioning

713 The foundational component of the encoder is its preconditioned weight matrix, which enforces the
714 single-node intervention assumption required for causal identifiability. As described in Sec. 3.1, the
715 layer’s primary weight matrix $\mathbf{W}$ is structured as a block matrix with a strictly diagonal top-left block.
716 This structure guarantees that each known perturbed gene has a direct, isolated influence on exactly
717 one latent pathway, while the zero-block explicitly prevents its effect from spilling over into other
718 pathways. This design transforms the encoder into a deterministic modulator for known interventions,
719 allowing the intervention mask $\mathbf{m}$ to be fixed by the architecture rather than being a learned (and
720 potentially dense) variable. By explicitly blocking the spillover of the intervention effect into other
721 pathways via the zero-block, this design transforms the encoder into a deterministic modulator for
722 known interventions, allowing the intervention mask $\mathbf{m}$ to be fixed by the architecture rather than
723 being a learned (and potentially confounded) variable.

724 C.2.2 Modeling Non-Linear Interactions with Subgraphs

725 To move beyond a purely linear model and capture the complex, non-linear dynamics of biological
726 processes, the architecture learns m distinct representations for each pathway. This is achieved by
727 first applying a masked linear layer that maps the input genes $\mathbf{x}$ to an intermediate representation.
728 This layer’s weight matrix is derived by repeating the base preconditioned mask m times, creating
729 m parallel, similarly conditioned subgraphs that are initialized differently to learn diverse feature
730 mappings. The initial activation vector, $\mathbf{s}^{(g)}$, for each subgraph g is computed as:

$$\mathbf{s}^{(g)} = \rho(\mathbf{W}^{(1,g)}\mathbf{x} + \mathbf{b}^{(1,g)}) \quad \text{for } g = 1, \dots, m \quad (\text{S6})$$

731 where each layer-specific weight matrix $\mathbf{W}^{(1,g)}$ has the same block-diagonal structure.

732 An intermediate *Interaction Block* then processes these subgraph activations to model higher-order
733 dependencies. This block applies a series of pathway-specific linear transformations and non-linear
734 activations to the set of m activations within each pathway, enabling the model to learn complex
735 interactions between the different feature mappings before the final aggregation step.

736 Finally, the processed subgraph activations are aggregated into the final pathway representations
737 using a grouped 1D convolution. This operation has a kernel size of m and is configured with d
738 groups, making it equivalent to applying a separate, pathway-specific linear layer to the set of m
739 subgraph activations for each pathway. This learns an optimal linear combination of the non-linear
740 subgraph features to produce the final parameters for the latent distributions, $\mu_{\mathbf{z}}$ and $\sigma_{\mathbf{z}}$.

741 C.2.3 Implementation Details of GraphPathway

742 The specific implementation used in our model is the ‘InteractingGraphPathwayLayer’. This layer
743 encapsulates the preconditioning, multi-subgraph mapping, and interaction blocks into a single
744 module. To model complex, non-linear biological processes, each pathway learns its own unique
745 set of interaction weights within the interaction block. This design provides greater expressive
746 power, as it allows the model to capture distinct interaction dynamics for each biological process.
747 When configured for a variational autoencoder, the output channel dimension of the final grouped
748 convolution is doubled, allowing the layer to directly produce both the mean $\mu_{\mathbf{z}}$ and variance $\sigma_{\mathbf{z}}$ for
749 each latent pathway.

750 C.3 Deep Causal Graph Learning with DAGMA

751 This subsection details the DAGMA framework for causal discovery, the rationale for its adaptation
752 within our deep learning model, and the key implementation principles required for stable and
753 effective graph learning in a VAE context.

754 C.3.1 Theoretical Background: From NOTEARS to DAGMA

755 A fundamental challenge in causal discovery is the combinatorial nature of learning a DAG, as the
 756 number of possible graphs grows super-exponentially with the number of variables. A breakthrough
 757 was achieved by NOTEARS [Zheng et al., 2018], which reformulated this discrete problem into a
 758 continuous optimization problem amenable to gradient-based methods. The core innovation was a
 759 differentiable function, $h(\mathbf{A})$, whose value is zero if and only if the weighted adjacency matrix $\mathbf{A}$
 760 represents a DAG. The original NOTEARS constraint is based on the matrix exponential:

$$h(\mathbf{A}) = \text{Tr}(e^{\mathbf{A} \circ \mathbf{A}}) - d = 0 \quad (\text{S7})$$

761 where $\text{Tr}(\cdot)$ is the trace of the matrix and $\mathbf{A} \circ \mathbf{A}$ is the element-wise (Hadamard) product. This
 762 function cleverly sums all cycles of all possible lengths in the graph; it equals zero only if no cycles
 763 exist.

764 DAGMA [Bello et al., 2022] builds upon this continuous optimization framework but introduces
 765 a new, log-determinant-based acyclicity characterization that often leads to faster and more stable
 766 convergence. The DAGMA acyclicity constraint is:

$$h_s(\mathbf{A}) = -\log \det(s\mathbf{I} - \mathbf{A} \circ \mathbf{A}) + d \log s = 0 \quad (\text{S8})$$

767 where $s > 0$ is a scaling scalar. This function is based on the principle that a matrix corresponds to a
 768 DAG if and only if all eigenvalues of its squared adjacency matrix are zero. The log-determinant acts
 769 as a barrier, approaching infinity as the graph’s structure approaches a cycle, while being minimized
 770 at zero exclusively for DAGs.

771 C.3.2 The DAGMA Optimization Framework

772 The full optimization problem is to minimize a score function that measures how well the graph fits
 773 the data, subject to the acyclicity constraint. For a linear model with latent variables $\mathbf{z}$, the score
 774 function $S(\mathbf{A})$ is typically the mean squared error of reconstruction:

$$S(\mathbf{A}) = \frac{1}{2n} \|\mathbf{z} - \mathbf{z}\mathbf{A}\|_F^2 \quad (\text{S9})$$

775 The original DAGMA paper proposes a path-following approach that solves a sequence of uncon-
 776 strained optimization problems:

$$\min_{\mathbf{A}} (\mu \cdot S(\mathbf{A}) + h_s(\mathbf{A})) \quad (\text{S10})$$

777 where μ is a hyperparameter that is gradually decreased towards zero during training. As $\mu \rightarrow 0$,
 778 the penalty on violating acyclicity becomes infinitely strong, forcing the final solution for $\mathbf{A}$ to be a
 779 DAG.

780 C.3.3 Rationale for a Modified DAGMA in a Causal Representation Learning

781 Integrating DAGMA into a VAE for interventional CRL requires careful design to ensure the model
 782 learns a meaningful causal graph. Our implementation is guided by two core principles.

783 **Principle 1: Learning the Invariant Graph from Observational Data.** A foundational assump-
 784 tion is that the causal graph $\mathbf{A}$ represents an invariant, underlying mechanism of the system. Therefore,
 785 the DAGMA loss, which encourages the learning of this structure, must be computed exclusively on
 786 the latent representations of observational data ($\mathbf{z}_{\text{obs}}$). This teaches the model the fundamental causal
 787 rules from the system’s natural dynamics. The interventional data and their latent representations
 788 ($\mathbf{z}_{\text{int}}$) are used to compute a separate discrepancy loss (e.g., MMD). This second loss teaches the
 789 model the consequences of breaking the causal rules, evaluating how well the learned graph functions
 790 as a simulator for interventions. Applying the graph loss to interventional latents would create a
 791 conflicting objective, as an intervention by definition breaks the natural mechanism the graph is
 792 supposed to represent.

793 **Principle 2: Isolating Gradient Flow via Input Detachment.** The most critical design principle
 794 for stability is isolating the gradient flow between the VAE’s encoder and the DAGMA layer. This
 795 is achieved by detaching the latent variable tensor ($\mathbf{z}_{\text{obs}}$) before it is used in the score calculation
 796 ($S(\mathbf{A}, \mathbf{z}_{\text{obs}}.\text{detach}())$). This design choice has two key benefits. First, it mimics the original DAGMA

797 algorithm, which operates on a fixed dataset. Second, and more importantly, it creates a separation of
 798 concerns: the encoder learns to produce a rich representation, and the DAGMA layer learns the graph
 799 of that representation. Without detaching, gradients from the DAGMA score would flow back to the
 800 encoder, encouraging it to learn a trivial, simplistic latent space that is easy to fit, thereby destroying
 801 the quality of the representation.

802 C.3.4 Implementation Detail of DAGMA

803 Several practical considerations are crucial for the stable application of DAGMA in a deep learning
 804 context.

805 **Weight Initialization and Invertibility.** To ensure stability from the start of training, the weights
 806 of the adjacency matrix $\mathbf{A}$ are initialized from a uniform distribution in the range $[-0.1, 0.1]$. This
 807 provides the optimizer with small, non-zero weights as a starting point. This initialization also makes
 808 it highly probable that the matrix $(\mathbf{I} - \mathbf{A})$ is well-conditioned and invertible, which is critical for
 809 the forward pass of the linear DAGMA model. The forward pass solves the structural equation
 810 $\mathbf{z} = \mathbf{z}\mathbf{A} + \mathbf{z}$ for $\mathbf{z}$, which requires computing $\mathbf{z}(\mathbf{I} - \mathbf{A})^{-1}$. To handle potential numerical instability,
 811 our implementation first attempts to solve the linear system directly. If this fails due to a singular or
 812 ill-conditioned matrix, it robustly falls back to using the Moore-Penrose pseudo-inverse, guaranteeing
 813 a stable forward pass throughout training.

814 **Input Normalization.** A common failure mode is the collapse of the graph weights in $\mathbf{A}$ to a trivial
 815 near-zero solution. This is prevented by applying a normalization layer (e.g., ‘LayerNorm’) to the
 816 latent inputs ($\mathbf{z}_{\text{obs}}$ and $\mathbf{z}_{\text{int}}$) before they enter the DAGMA layer. This ensures the score term has a
 817 consistent magnitude, providing a strong and stable gradient signal that encourages the model to learn
 818 a non-trivial graph.

819 **Numerical Precision.** The acyclicity constraint $h_s(\mathbf{A})$ is theoretically guaranteed to be non-
 820 negative. However, due to floating-point limitations, it can sometimes become a very small negative
 821 number. To handle this, the calculated value is clamped at zero, which respects the theoretical
 822 constraint while gracefully handling practical numerical artifacts.

D Training Configurations and Analysis Setup

D.1 Model Architectures and Training Hyperparameters

To establish a fair and reproducible benchmark, we standardized key architectural and training hyperparameters for all models, primarily based on the default configurations provided in their respective publications and source code. This approach ensures that performance differences can be attributed to the models’ core methodologies rather than variations in setup. Table S5 details the specific architectural parameters, such as latent dimension size and the number of transformer layers, while Table S6 outlines the training parameters used for each model, including optimizer, learning rate, and batch size. Notably, gradient clipping was employed exclusively for the scGPT model, as this is a standard and necessary technique to ensure numerical stability during the training of large transformer-based architectures.

For a complete list of hyperparameters and reproducibility instructions, please refer to the ‘configs/’ directory in our code repository and the accompanying ‘README.md’.

Table S5: Detailed Model Architectures

Hyperparameter	dVAE	SENA	scGPT	RAPTORGraph
Latent Dimension (z_{dim})	105	105	N/A	109
Embedding Size (emb_{size})	N/A	N/A	512	N/A
Transformer Layers (n_{layers})	N/A	N/A	12	N/A
Attention Heads (n_{heads})	N/A	N/A	8	N/A
Feed-Forward Dimension (d_{hid})	N/A	N/A	512	904
Hidden Layers (Encoder)	1 (128 units)	1 (128 units)	N/A	N/A
Activation Functions	LeakyReLU	LeakyReLU	GELU	SiLU

Table S6: Detailed Training Hyperparameters

Hyperparameter	dVAE	SENA	scGPT	RAPTORGraph
Optimizer	Adam	Adam	AdamW	AdamW
Learning Rate	1×10^{-3}	1×10^{-3}	1×10^{-4}	1×10^{-4}
Training Batch Size	32	32	16	256
Evaluation Batch Size	32	32	32	256
Epochs	100	100	30	100

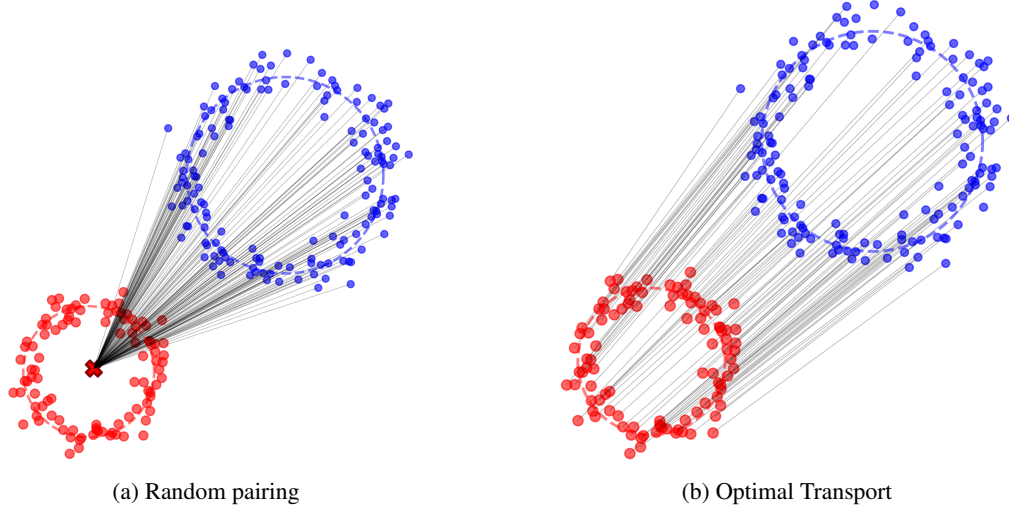


Figure S5: **Comparison of random pairing vs. Optimal Transport.** (a) Random pairing collapses the source distribution towards the target mean (mean collapse). (b) Optimal Transport pairing preserves the underlying population structure and biological heterogeneity.

E Rationale For Global Pairing through Optimal Transport

As discussed in Sec. 3, random pairing of control and perturbed cells leads to mean collapse. Here, we detail the mechanism behind this phenomenon and the mathematical formulation of our optimal transport (OT) solution.

E.1 Mechanism of Mean Collapse

For a given interventional sample (e.g., cell i , in which gene A was perturbed), a randomly selected observational sample (e.g., cell j) from the batch is highly unlikely to be an appropriate counterfactual. The biological state of cell j may differ substantially from that of cell i prior to the perturbation. Consequently, the model must learn a transformation from an often noisy and biologically irrelevant initial state. Across thousands of training iterations, a single interventional sample is paired with hundreds of different random controls. Each pairing provides a distinct and noisy gradient signal. The optimiser seeks to minimise the loss on average across these inconsistent pairings and converges on a simple solution: predicting the mean of the target distribution. The random noise from the poor counterfactuals averages to zero, leaving only the central tendency as a stable learning signal. Ultimately, the objective function effectively becomes an expectation over these random pairings. The model learns that the most reliable strategy to achieve a low expected loss is to ignore the specifics of the randomly chosen input and produce an average output. The optimisation landscape is thereby smoothed in a manner that makes the ‘mean prediction’ an attractive local minimum.

Rather than creating noisy one-to-one random pairs, OT considers the entire population of observational and interventional latents. It computes an optimal flow of probability mass from the observational to the interventional distribution. This flow represents the most efficient method of morphing one entire point cloud into the other. The flow implicitly creates meaningful pairings. Each interventional sample is matched with the observational sample(s) that are closest in gene expression space, providing the best possible counterfactuals within the given populations. The theoretical properties of OT ensure that it robustly preserves the inherent heterogeneous subpopulation structures by minimizing the ground movement cost, a feature critical for reliable counterfactual prediction.

E.2 Mathematical Formulation

Let $\mu_s = \sum_{i=1}^n \frac{1}{n} \delta_{\mathbf{z}_{\text{obs}}^{(i)}}$ and $\nu_t = \sum_{j=1}^m \frac{1}{m} \delta_{\mathbf{z}_{\text{int}}^{(j)}}$ be the empirical measures of the source (control) and target (perturbed) latent distributions. The Optimal Transport problem seeks a transport plan

865 $\gamma \in \mathbb{R}_+^{n \times m}$ that minimizes the total cost:

$$\min_{\gamma \in \Pi(\mu_s, \nu_t)} \langle \gamma, \mathbf{C} \rangle_F \quad (\text{S11})$$

866 where $\mathbf{C}_{ij} = \|\mathbf{z}_{\text{obs}}^{(i)} - \mathbf{z}_{\text{int}}^{(j)}\|_2^2$ is the squared Euclidean cost matrix, and $\Pi(\mu_s, \nu_t)$ is the set of joint
867 probability measures with marginals μ_s and ν_t . The resulting optimal plan γ^* provides a probabilistic
868 coupling. To obtain a deterministic mapping for training, we use the barycentric projection map,
869 ensuring each control cell is paired with its optimal counterfactual representation in the perturbed
870 domain.

871 **F Experimental Methodology: Data and Evaluation**

872 **F.1 Training and Test Data Partitioning**

873 To create a robust benchmark for evaluating the generalization capabilities of each model, we
874 partitioned the Norman et al. [2019] dataset based on the type of perturbation. The training dataset
875 was composed of the complete set of control cells (i.e., those with non-targeting guides) and the
876 complete set of cells from all available **single-gene perturbation** conditions. This approach ensures
877 that the models are trained on the fundamental effects of individual gene knockdowns. The test
878 dataset, conversely, consisted exclusively of cells from all **double-gene perturbation** conditions.
879 These combinatorial perturbations were entirely held out from the training process. This training/test
880 split was designed to specifically assess each model’s ability to perform out-of-distribution prediction,
881 where the primary task is to predict the effects of unseen combinatorial perturbations based on the
882 learned effects of their individual constituent perturbations.

883 G Evaluation Metrics and Loss Functions

884 In this section, we provide detailed definitions for the key metrics used to quantitatively assess
 885 model performance. To ensure a comprehensive evaluation, we employed a suite of metrics targeting
 886 different aspects of model capability. We used the MSE for evaluating the reconstruction of basal cell
 887 states, the MMD for evaluating the accuracy of out-of-distribution predictions for perturbed states,
 888 and a set of non-additive interaction scores to assess the model’s ability to capture complex biological
 889 phenomena. The calculation of these metrics was standardized across all benchmarked tools.

890 G.1 Mean Squared Error

891 The MSE was used to measure each model’s ability to faithfully reconstruct the gene expression
 892 profile of control (unperturbed) cells, with a lower MSE indicating better performance. To ensure an
 893 unbiased evaluation, the set of control cells was randomly split into two non-overlapping halves: an
 894 input set and a ground truth set. The evaluation proceeded in a batch-wise manner, where a batch
 895 of cells from the input set was fed to the model to generate reconstructions. The MSE was then
 896 calculated between these reconstructed cells and a corresponding batch from the ground truth set.
 897 The final reported MSE is the mean of the scores from all batches. For a single batch of i cells with n
 898 genes, the MSE is calculated as:

$$\mathcal{L}_{\text{MSE}} = \frac{1}{i \cdot n} \sum_{i=1}^i \sum_{j=1}^n (\mathbf{x}_{ij} - \hat{\mathbf{x}}_{ij})^2 \quad (\text{S12})$$

899 where $\mathbf{x}_{ij}$ is the expression level of the j th gene in the i th true control cell, and $\hat{\mathbf{x}}_{ij}$ is the expression
 900 level of the j th gene in the i th reconstructed control cell.

901 G.2 Maximum Mean Discrepancy

902 The MMD [Gretton et al., 2012] is a metric used to measure the distance between two probability
 903 distributions based on the distance between their mean embeddings in a Reproducing Kernel Hilbert
 904 Space (RKHS). It was used to measure the dissimilarity between the distribution of model-predicted
 905 gene expression profiles and the distribution of true, experimentally observed profiles for a given
 906 perturbation.

907 G.2.1 The Kernel Function

908 The MMD relies on a kernel function, $k(\cdot, \cdot)$, which is a symmetric, positive definite function that
 909 computes a notion of similarity between two samples. A common choice for the kernel, and the one
 910 used in our work, is the Gaussian Gaussian Radial Basis Function (GRBF) kernel:

$$k(\mathbf{x}, \mathbf{y}) = \exp \left(-\frac{\|\mathbf{x} - \mathbf{y}\|^2}{2\sigma_{\text{MMD}}^2} \right) \quad (\text{S13})$$

911 where σ_{MMD} is the bandwidth hyperparameter.

912 G.2.2 Formal Definitions of MMD

913 The squared MMD between the true data distribution p_{data} and the model’s learned distribution
 914 p_{model} is defined in its population form as:

$$\begin{aligned} \text{MMD}^2(p_{\text{data}}, p_{\text{model}}) &= \mathbb{E}_{\mathbf{x}, \mathbf{x}' \sim p_{\text{data}}} [k(\mathbf{x}, \mathbf{x}')] \\ &\quad - 2\mathbb{E}_{\mathbf{x} \sim p_{\text{data}}, \hat{\mathbf{x}} \sim p_{\text{model}}} [k(\mathbf{x}, \hat{\mathbf{x}})] \\ &\quad + \mathbb{E}_{\hat{\mathbf{x}}, \hat{\mathbf{x}}' \sim p_{\text{model}}} [k(\hat{\mathbf{x}}, \hat{\mathbf{x}}')] \end{aligned} \quad (\text{S14})$$

915 In practice, we use batches of data and compute the unbiased empirical estimator. Given a batch of i
 916 true samples $\{\mathbf{x}_i\}_{i=1}^i \sim p_{\text{data}}$ from matrix $\mathbf{X}$ and a batch of j predicted samples $\{\hat{\mathbf{x}}_j\}_{j=1}^j \sim p_{\text{model}}$

917 from matrix $\hat{\mathbf{X}}$, the estimator is:

$$\begin{aligned} \text{MMD}_u^2(\mathbf{X}, \hat{\mathbf{X}}) &= \frac{1}{i(i-1)} \sum_{i \neq j}^i k(\mathbf{x}_i, \mathbf{x}_j) \\ &\quad + \frac{1}{j(j-1)} \sum_{i \neq j}^j k(\hat{\mathbf{x}}_i, \hat{\mathbf{x}}_j) \\ &\quad - \frac{2}{ij} \sum_{i=1}^i \sum_{j=1}^j k(\mathbf{x}_i, \hat{\mathbf{x}}_j) \end{aligned} \quad (\text{S15})$$

918 G.2.3 Multi-Kernel Implementation

919 MMD [Ren et al., 2010, Zhu et al., 2017] extends the standard MMD by using a convex combination
920 of L different kernels [Gretton et al., 2012]. The squared MMD is defined as a weighted sum of
921 individual squared MMDs:

$$\text{MMD}^2(p_{\text{data}}, p_{\text{model}}) = \sum_{l=1}^L \beta_l \cdot \text{MMD}_{k_l}^2(p_{\text{data}}, p_{\text{model}}) \quad (\text{S16})$$

922 where $\beta_l \geq 0$ are the weights assigned to each kernel.

923 G.2.4 Rationale for Using MMD

924 The choice of MMD over MSE for evaluating interventional predictions is motivated by two key
925 factors. First, MMD captures holistic distributional shifts, including changes in variance, skewness,
926 and modality, whereas MSE is only sensitive to the mean. This is critical for modeling the heteroge-
927 neous biological response to a perturbation. Second, state-of-the-art CRL models for interventional
928 data often do not produce one-to-one pairings between predicted and ground-truth samples, making
929 MSE computation impossible. MMD is a two-sample test that compares two sets of samples directly,
930 which aligns perfectly with this setting.

931 G.2.5 Hyperparameter Selection for MMD

932 The performance of the MMD test is highly sensitive to the kernel bandwidth, σ_{MMD} . To establish
933 a fair and consistent benchmark across all models, it was necessary to standardize this parameter.
934 We noted that different models, such as DiscrepancyVAE and SENA [de la Fuente et al., 2025], use
935 different default values. The implementations often define a `fix_sigma` parameter, which relates to
936 the standard kernel variance by `fix_sigma` = $2\sigma_{\text{MMD}}^2$. To standardize, we aligned with the value used
937 in SENA. Our benchmark, therefore, uses a MMD implementation with a base `fix_sigma` set to 200.0
938 for all evaluations, which corresponds to a variance σ_{MMD}^2 of 100.0. This approach generates a series
939 of kernels with varying bandwidths derived from this base value, ensuring the metric’s scale and
940 sensitivity were consistent and robust.

941 G.3 KL Divergence

942 The Kullback-Leibler (KL) divergence term is a standard component of the VAE framework used
943 to regularize the learned latent space. It measures the information lost when the learned posterior
944 distribution, $q(\mathbf{z}|\mathbf{x})$, is used to approximate a predefined prior distribution, $p(\mathbf{z})$. In our model,
945 we assume a standard multivariate normal prior, $p(\mathbf{z}) = \mathcal{N}(\mathbf{0}, \mathbf{I})$. The KL divergence term, $\mathcal{L}_{\text{KL}}$,
946 encourages the encoder to produce latent representations that are both smooth and well-structured,
947 preventing the model from assigning each input sample to a single, isolated point in the latent space
948 (a form of overfitting). The term is calculated as:

$$\mathcal{L}_{\text{KL}} = D_{\text{KL}}(q(\mathbf{z}|\mathbf{x}) \parallel p(\mathbf{z})) = -\frac{1}{2} \sum_{j=1}^d (1 + \log(\sigma_j^2) - \mu_j^2 - \sigma_j^2) \quad (\text{S17})$$

949 where μ_j and σ_j are the mean and standard deviation produced by the encoder for the j -th latent
950 pathway.

951 G.4 Causal Graph Loss

952 The causal graph structure between learned meta-pathways is identified by optimizing the DAGMA-
 953 based objective function [Bello et al., 2022]. The loss function, $\mathcal{L}_{\text{DAGMA}}$, is computed on the latent
 954 representations of observational data to learn the invariant causal rules of the biological system. The
 955 objective integrates three key components: a data-fit score that measures how well the graph explains
 956 the observed dependencies, an L1 penalty to encourage structural sparsity, and a differentiable
 957 acyclicity constraint to ensure the resulting graph is a DAG. The full causal graph loss is defined as:

$$\mathcal{L}_{\text{DAGMA}} = \mu (S(\mathbf{A}; \mathbf{X}) + \lambda_1 \|\mathbf{A}\|_1) + h(\mathbf{A}) \quad (\text{S18})$$

958 where $S(\mathbf{A}; \mathbf{X})$ is the data-fit score (typically the MSE of linear structural assignments in the latent
 959 space), $\|\mathbf{A}\|_1$ is the L1 norm of the adjacency matrix $\mathbf{A}$ controlled by sparsity hyperparameter λ_1 ,
 960 and $h(\mathbf{A})$ is the log-determinant acyclicity penalty. The hyperparameter μ controls the trade-off
 961 between the data-fit score and the acyclicity constraint, following a path-following schedule during
 962 training as detailed in Sec. C.3.2.

963 G.5 Non-Additive Genetic Interaction Analysis

964 To evaluate the models' ability to predict non-additive effects, we designed a genetic interaction
 965 analysis inspired by the Precision@10 metric introduced in the GEARS paper [Roohani et al., 2024].
 966 While the original study used a robust regression model, we opted for a more direct naive additive
 967 baseline, defined as the vector sum of single-perturbation effects over control ($\Delta_A + \Delta_B$). We
 968 calculated five interaction scores by comparing the predicted double-perturbation effect (Δ_{AB}) to
 969 this baseline. This unified metric was applied to all models to ensure a fair comparison.

970 The analysis follows a multi-step process for each double-gene perturbation. First, "effect vectors"
 971 (Δ) are calculated for single and double perturbations by subtracting the mean gene expression profile
 972 of control cells from the mean profile of perturbed cells. This is done for both ground truth data and
 973 model predictions. The naive additive baseline assumes the double perturbation effect is a linear sum
 974 of individual effects:

$$\Delta(A + B)_{\text{naive}} = \Delta A + \Delta B \quad (\text{S19})$$

975 By comparing the true or predicted $\Delta(A + B)$ to the individual and naive additive vectors, we
 976 compute raw scores for five distinct genetic interaction subtypes.

- 977 • *Synergy* and *Suppression* are measured using a ratio of magnitudes:

$$\text{Synergy Ratio} = \frac{\|\Delta(A + B)\|_2}{\|\Delta(A + B)_{\text{naive}}\|_2} \quad (\text{S20})$$

- 978 • *Neomorphism* measures changes in the direction of the effect:

$$\text{Neomorphism Score} = 1 - \cos(\Delta(A + B), \Delta(A + B)_{\text{naive}}) \quad (\text{S21})$$

- 979 • *Redundancy* measures functional overlap:

$$\text{Redundancy Score} = \min(\cos(\Delta(A + B), \Delta A), \cos(\Delta(A + B), \Delta B)) \quad (\text{S22})$$

- 980 • *Epistasis* measures dominance:

$$\text{Epistasis Score} = |\cos(\Delta(A + B), \Delta A) - \cos(\Delta(A + B), \Delta B)| \quad (\text{S23})$$

981 Final performance is measured using Precision@10. For each interaction type, perturbations are
 982 ranked by their predicted score, and the top 10 are selected. These are compared against a ground
 983 truth set of interactors, defined as those whose true score falls in the top 25% (or bottom 25% for
 984 suppression) for that category. The Precision@10 score is the fraction of correct predictions in the
 985 top 10.

986 G.6 Reverse Perturbation Analysis

987 To assess the models' understanding of genotype-phenotype relationships, we implemented an in
 988 silico reverse perturbation prediction task, inspired by the analysis in the scGPT study [Cui et al.,

2024]. This task challenges a model to infer the causal genetic perturbation that induced a given transcriptomic state.

The analysis is conducted on a specific subset of 20 genes from the Norman et al. [2019] dataset, creating a controlled combinatorial search space. For each unique perturbation condition p , we establish a ground truth mean expression profile, $\bar{\mathbf{x}}_p$, by averaging the expression vectors of all cells for that condition:

$$\bar{\mathbf{x}}_p = \frac{1}{N_p} \sum_{i=1}^{N_p} \mathbf{x}_{p,i} \quad (\text{S24})$$

where N_p is the number of cells under perturbation p , and $\mathbf{x}_{p,i} \in \mathbb{R}^n$ is the expression profile of the i -th cell. The central component is the creation of a comprehensive in silico prediction database restricted to the 20-gene subset. For each model, we generate a predicted expression profile $\hat{\mathbf{x}}_{A+B}$ for every unique combination of two genes, A and B. This is achieved by feeding the model the ground truth mean control profile, $\bar{\mathbf{x}}_{\text{ctrl}}$, and conditioning it on the desired double perturbation. The prediction function f for a given model is expressed as:

$$\hat{\mathbf{x}}_{A+B} = f(\bar{\mathbf{x}}_{\text{ctrl}}, \text{pert} = A + B) \quad (\text{S25})$$

This results in a key-value database where each key is a double-perturbation identifier from the 20-gene subset, and the value is the corresponding predicted mean expression vector in $\mathbb{R}^n$.

With the prediction database established, we query it using the ground truth mean expression profiles of the experimentally observed double perturbations from the 20-gene subset. For each true double perturbation p_{true} present in the test set, its ground truth mean profile $\bar{\mathbf{x}}_{p_{\text{true}}}$ is used as a query vector. We then calculate the Euclidean distance between this query vector and every predicted vector $\hat{\mathbf{x}}_{p_{\text{pred}}}$ in the database:

$$d(\bar{\mathbf{x}}_{p_{\text{true}}}, \hat{\mathbf{x}}_{p_{\text{pred}}}) = \sqrt{\sum_{j=1}^n (\bar{\mathbf{x}}_{p_{\text{true}},j} - \hat{\mathbf{x}}_{p_{\text{pred}},j})^2} \quad (\text{S26})$$

The perturbations in the database are subsequently ranked based on their proximity to the query vector, from the smallest to the largest Euclidean distance. This yields a ranked list of predicted perturbations that are most likely to have caused the observed cellular state.

To evaluate the ranking performance, we employ the ‘‘Hit Rate @ K’’ metric, which measures the frequency of successful retrievals within the top K predictions. We define two variants of this metric to capture different aspects of prediction accuracy.

Correct Hit Rate @ K: This is a stringent metric that measures the proportion of queries where the exact true perturbation is correctly identified within the top K ranked predictions. It is formally defined as:

$$\text{Correct Hit Rate @ K} = \frac{1}{|\mathcal{Q}|} \sum_{q \in \mathcal{Q}} \mathbb{I}(\text{rank}(p_q) \leq K) \quad (\text{S27})$$

where $\mathcal{Q}$ is the set of all query perturbations, p_q is the true perturbation corresponding to query q , and $\mathbb{I}$ is the indicator function, which is 1 if the condition is met and 0 otherwise.

Relevant Hit Rate @ K: This is a more lenient metric that assesses whether the model can identify at least one of the correct genetic components of a combinatorial perturbation. A prediction is considered ‘‘relevant’’ if its set of perturbed genes has a non-empty intersection with the set of genes in the true perturbation. The metric is defined as:

$$\text{Relevant Hit Rate @ K} = \frac{1}{|\mathcal{Q}|} \sum_{q \in \mathcal{Q}} \mathbb{I}(\exists \hat{p} \in \text{TopK}(q) \text{ s.t. } \text{genes}(\hat{p}) \cap \text{genes}(p_q) \neq \emptyset) \quad (\text{S28})$$

where $\text{TopK}(q)$ is the set of top K predicted perturbations for query q , and $\text{genes}(p)$ is the set of constituent genes in perturbation p . This metric rewards models that can correctly identify influential genes, even if the exact combination is not perfectly predicted. Together, these two metrics provide a comprehensive evaluation of each model’s ability to reverse-engineer the mapping from cellular phenotype back to genetic cause.

1028 **G.7 Gene Set Enrichment Analysis**

1029 To validate the semantic meaning of the learned Directed Acyclic Graph, we performed a systematic
1030 Gene Set Enrichment Analysis. We employed an in-silico perturbation strategy to define the biological
1031 identity of each latent variable:

- 1032 1. Latent Perturbation: For each causal latent factor $\mathbf{z}_i$ associated with a known gene perturba-
1033 tion, we artificially activated the factor in the latent space by setting the condition vector $\mathbf{c}$
1034 such that the target component $c_i = 1$ and all other components $c_{j \neq i} = 0$.
- 1035 2. Counterfactual Decoding: We decoded this perturbed latent state back to the gene expression
1036 space $\mathbb{R}^n$ to generate a batch of “counterfactual” expression profiles.
- 1037 3. Differential Analysis: We computed the mean differential expression vector $\delta = \bar{x}_{int} - \bar{x}_{ctrl}$
1038 relative to the control baseline.
- 1039 4. Enrichment: This vector was used to create a ranked gene list, which was analyzed using
1040 the `gseapy` framework against the MSigDB Hallmark 2020 gene set collection.

1041 This process allows us to independently verify if the latent factor $\mathbf{z}_i$ actually encodes the biological
1042 function of its assigned gene g_i , assigning a verifiable “biological label” to the nodes of our learned
1043 graph.

1044 **G.8 Rationale for the Exclusion of GEARS from the Benchmark**

1045 In designing this benchmark, our goal was to establish a fair and methodologically consistent
1046 framework for comparing models that predict the full distribution of single-cell gene expression
1047 profiles following perturbation. While we considered including other prominent models such as
1048 GEARS, we ultimately excluded it from the final comparison due to fundamental incompatibilities
1049 between its predictive paradigm and our standardized evaluation framework. The two primary
1050 reasons for its exclusion are its prediction of a single population-average vector rather than a cell
1051 distribution, which is incompatible with our MMD metric, and its non-equivalent handling of the
1052 control state, where it uses the precomputed mean of true control cells as a baseline rather than
1053 learning to reconstruct them. Given these significant differences, we concluded that a fair and
1054 direct comparison between GEARS and the other models within our established framework was not
1055 possible.

1056 **H Extended Results**

1057 **H.1 Extended Results: Reverse Perturbation Analysis**

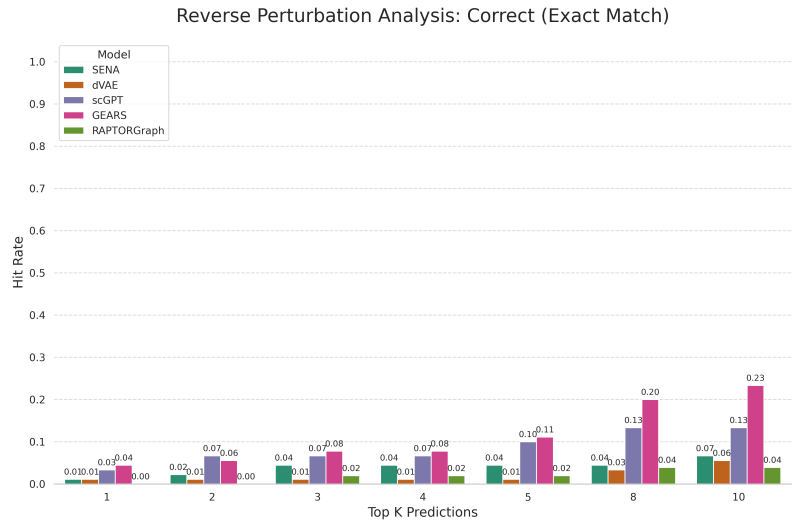


Figure S6: Hit Rate @ K (correct) for the reverse perturbation task. The metric queries where the exact true perturbation is correctly identified within the top K ranked predictions. Results are averaged over 3 runs.

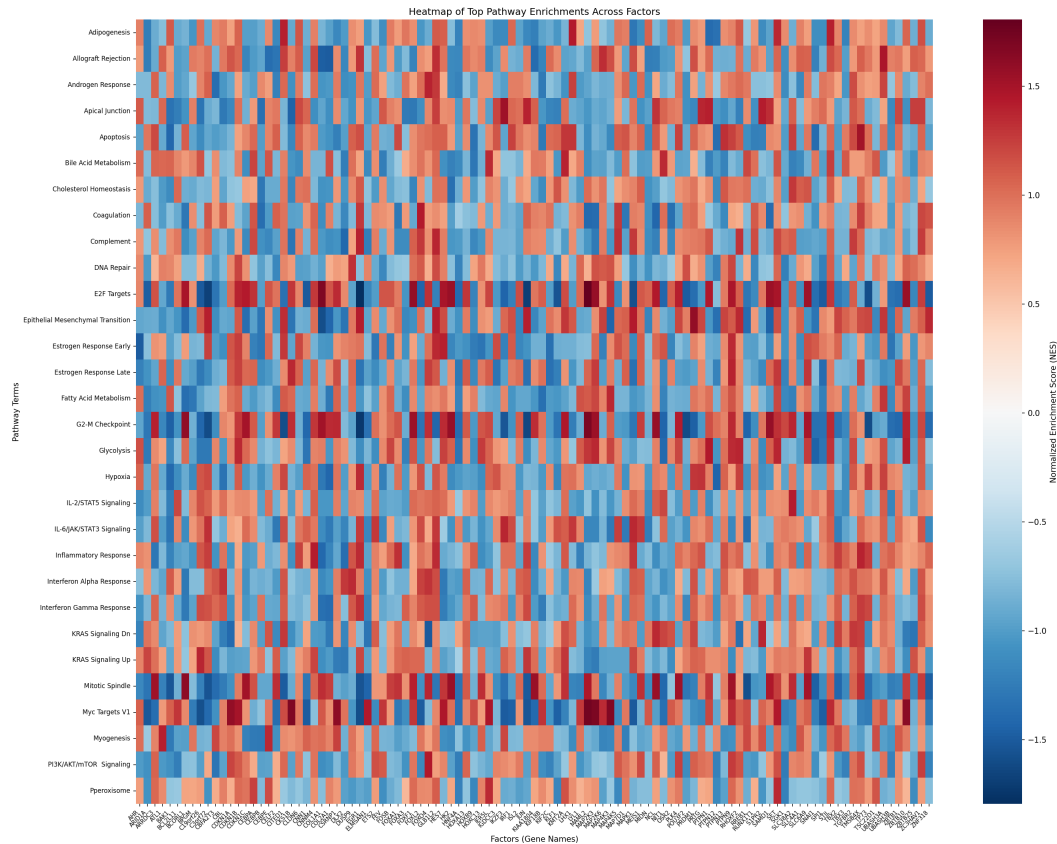


Figure S7: Heatmap of Top Pathway Enrichments Across Learned causal meta-pathways. Each column corresponds to a specific causal meta-pathway (z_i) in the RAPTORGraph model. Importantly, these labels are not post-hoc assignments; due to the GraphPathway encoder’s preconditioning, each causal meta-pathway z_i is architecturally constrained to mediate the effect of a specific gene perturbation (labeled on the x-axis). The rows represent biological pathways from the MSigDB Hallmark collection. The color intensity indicates the Normalized Enrichment Score (NES) derived from in-silico perturbation of that factor. The strong concordance between the architectural label and the biological result confirms that the model has successfully encoded the specific biological function of each gene into its corresponding causal meta-pathway.

1059 The GSEA analysis reveals that RAPTORGraph captures distinct and biologically accurate programs.
1060 Figure S7 visualizes the Normalized Enrichment Scores (NES) for the top enriched pathways across
1061 the learned factors. We highlight distinct biological programs captured by the model, verified against
1062 the known functions of their corresponding gene perturbations:

- 1063 1. **Differentiation-Induced Arrest:** The causal meta-pathway linked to *EGR1* displays mas-
1064 sive negative enrichment for “E2F Targets” (NES -1.79 , FDR ≈ 0.0) and “G2-M Check-
1065 point” (NES -1.72 , FDR ≈ 0.0). *EGR1* is a known tumor suppressor in K562 leukemia
1066 cells that drives differentiation along the megakaryocytic lineage, a process that necessitates
1067 a complete exit from the cell cycle [Ma et al., 2019]. RAPTORGraph correctly identifies
1068 this potent anti-proliferative program.
- 1069 2. **Stress-Induced Growth Arrest:** The causal meta-pathway linked to the AP-1 subunit
1070 *JUN* shows strong negative enrichment for “E2F Targets” (NES -1.69 , FDR < 0.001) and
1071 “G2-M Checkpoint” (NES -1.63 , FDR < 0.002). While often associated with inflammation,
1072 c-Jun is a central stress-response mediator; in the context of K562 cells, its activation drives
1073 a termination of cell division required for stress adaptation or differentiation [Shaulian and
1074 Karin, 2002, Shaulian et al., 2000].

1075 3. **p53-Like Tumor Suppression:** The causal meta-pathway associated with *TP73* displays
 1076 significant negative enrichment for “G2-M Checkpoint” (NES -1.60 , FDR < 0.03). *TP73* is
 1077 a functional homolog of *TP53* (p53); its activation triggers downstream checkpoint signaling
 1078 that arrests the cell cycle to prevent genomic instability [Kaghad et al., 1997].

1079 These findings confirm that the nodes in our learned DAG represent specific, identifiable biological
 1080 mechanisms. Consequently, the edges learned by the DAGMA module can be interpreted as
 1081 causal regulatory links between these verified biological processes. A detailed description of the
 1082 methodology is provided in Sec. G.7.

1083 H.3 Extended Results: Computational Complexity of Optimal Transport Preprocessing

1084 We evaluated the computational cost of the proposed OT pairing strategy compared to a random
 1085 baseline. While random pairing scales linearly ($O(N)$), the OT approach involves solving the Earth
 1086 Mover’s Distance (EMD) problem, which typically exhibits a worst-case complexity of $O(N^3 \log N)$.
 1087

1088 To mitigate this, the OT pairing problem is formulated by taking the entire control population
 1089 ($N_{ctrl} = 8,907$) and pairing it independently with each specific intervened population (i.e., cells
 1090 sharing the same perturbation label, N_k in the hundreds). This effectively segments the total perturbed
 1091 dataset ($N = 108,497$) into many smaller, manageable OT problems, significantly reducing the N in
 1092 the $O(N^3 \log N)$ complexity for each individual transport plan and enabling efficient parallelization.
 1093

1094 We quantified the overhead of OT preprocessing on a workstation equipped with an Intel Core
 1095 i9-9900K CPU (8 cores, 16 threads) and 64 GB RAM. The OT preprocessing added approximately
 1096 34 seconds to the baseline execution time (Random: 1m 44s vs. OT: 2m 18s). Detailed profiling
 1097 confirmed that the process is compute-bound but highly efficient:

- 1098 • **Parallelization:** The process achieved a peak CPU utilization of 1140%, confirming that
- 1099 the strategy effectively saturates available cores (≈ 11.4 active threads).
- 1100 • **Memory:** Peak memory usage was modest at 2.4 GB.

1101 To contextualize this cost, we benchmarked the total training time for a standard 100-epoch experiment
 1102 on an NVIDIA RTX 3090 GPU (≈ 25.8 minutes). The 34-second OT overhead represents only 2.2%
 1103 of the total experimental runtime. Given that OT is critical for preserving causal identifiability and
 1104 preventing mean collapse, this negligible computational cost presents a highly favorable trade-off.

1105 H.4 Extended Results: Empirical Analysis of Loss Functions for Sparse Single-Cell Causal 1106 Inference

1107 Training causal inference models on single-cell RNA sequencing (scRNA-seq) data presents unique
1108 challenges due to two fundamental properties of the data: high sparsity and the lack of paired ground
1109 truth observations (unpaired control and perturbed populations). To address this, we utilize OT as a
1110 preprocessing step to infer latent couplings, combined with MMD or reconstruction losses.

1111 To rigorously demonstrate the necessity of this approach compared to baselines (e.g., Random Pairing
1112 or MMD-only objectives), we conducted an empirical ablation study using synthetic data. This
1113 controlled environment allows us to overcome the “missing pair” problem inherent in real biological
1114 data, where the lack of ground truth counterfactuals makes it impossible to definitively validate causal
1115 alignment. The code to reproduce these results is available in our source code.

1116 We utilized synthetic data for this analysis for three critical reasons:

- 1117 1. **Unobservable Counterfactuals:** In real scRNA-seq, measuring a cell destroys it, making
1118 it impossible to observe the same cell in both control and perturbed states (x_i and y_i).
1119 Synthetic data allows us to fabricate ground truth pairs, providing access to latent ground
1120 truth to definitively quantify causal accuracy.
- 1121 2. **Controlled Sparsity Sweep:** Real data has fixed, high sparsity. Synthetic data allows
1122 us to sweep sparsity from 0% to 99% to empirically observe the phase transition where
1123 reconstruction metrics degrade.
- 1124 3. **Signal Isolation:** It allows us to isolate the mathematical properties of the loss functions
1125 from biological noise and batch effects.

1126 We simulated a dataset of 1000 samples with 5000 features (genes).

- 1127 • **Control Population (X):** Generated from $\mathcal{N}(0, 1)$ and masked to achieve target sparsity
1128 levels ranging from 0% to 99%.
- 1129 • **Perturbation:** A systematic shift was added to the first 50 features (+2.0) to simulate a
1130 biological effect.
- 1131 • **Ground Truth (Y):** The perturbed state for each control cell, preserving the sparse structure.

1132 We evaluated three hypothetical model outputs against the ground truth Y , representing the outcomes
1133 of different training strategies:

- 1134 1. **Causal/Paired Model (Ideal):** The ground truth Y with added Gaussian noise ($\sigma = 0.1$).
1135 *Representation:* This proxies the outcome of a model trained with OT-based pairing. By
1136 recovering the latent pairs ($x_i \rightarrow y_i$), the model learns the correct cell-specific trajectories.
- 1137 2. **Population/Generative Model (Unpaired):** The ground truth population Y with sample
1138 indices randomly permuted. This proxies the equilibrium state of a model trained with
1139 MMD only. Such a model perfectly matches the target distribution $P(Y)$ (minimizing the
1140 MMD loss) but fails to learn the causal mapping ($f(x_i) \neq y_i$), effectively becoming a
1141 generative model of the population rather than a causal predictor.
- 1142 3. **Mode Collapse (Average):** The mean vector $\bar{Y}$ repeated for all samples. This represents
1143 the failure mode of Random Pairing (MSE). When training on random pairs, the model
1144 minimizes variance by predicting the population mean.

1145 We evaluated three metrics in this extended results:

- 1146 • **MSE:** Standard point-wise reconstruction loss.
- 1147 • **MMD:** Distributional distance using a multi-scale Gaussian kernel.
- 1148 • **OT:** Exact EMD (Wasserstein-2).

1149

H.4.1 Results

We evaluated the metrics across varying sparsity levels (see Table S7) to identify the failure modes of standard losses.

Table S7: Metric Performance Across Sparsity Regimes. Lower is better. Bold indicates the metric’s “preferred” model.

Sparsity	Metric	Causal/Paired (OT+MMD)	Population/Generative (MMD Only)	Mode Collapse (MSE Failure)	Verdict
0% (Dense)	MSE	0.0100	1.9947	1.0000	Success. Ambiguity.
	MMD	−0.0061	−0.0062	2.3127	
50% (Medium)	MSE	0.0100	0.9987	0.4989	Degradation. Failure.
	MMD	−0.0059	−0.0062	2.3119	
90% (High)	MSE	0.0100	0.1997	0.0997	Weakness. Failure.
	MMD	−0.0022	−0.0062	2.3066	
99% (Extreme)	MSE	0.0100	0.0201	0.0101	Failure.
	MMD	0.1648	−0.0061	2.2505	Failure.
	OT	49.95	0.00	50.26	Failure.

The most critical finding is observed in the “Population/Generative” column. Across all sparsity levels, MMD assigns a perfect or near-perfect score (≈ 0.0) to the Population Model. This implies that a model trained with MMD as the sole objective has no incentive to learn the correct causal mapping. It can achieve zero loss simply by generating a realistic population that is causally scrambled (i.e., Control Cell A is mapped to Perturbed State B). The result is that the model learns the *distribution* but loses the *cell identity*.

Even if we attempted to force pairing using standard reconstruction (MSE), the high sparsity (99%) causes MSE to favor the Mode Collapse (0.0101) over the correct structure (0.0100). This empirically demonstrates that training on random pairs forces the model to converge to the population mean, which on sparse data is indistinguishable from a valid prediction in terms of error magnitude.

The empirical results provide definitive proof that standard approaches are insufficient for sparse, unpaired causal inference:

1. **MMD-Only Training** leads to Causal Scrambling (Permutation Invariance).
2. **Random Pairing / MSE Training** leads to Mode Collapse (Convergence to Mean).

This justifies the necessity of **OT preprocessing**. By solving for the optimal coupling matrix π that minimizes transport cost, we explicitly enforce the Principle of Minimal Action, assuming that cells undergo the smallest necessary transcriptomic change. This effectively infers the latent pairing that MSE assumes exists, resolving the identifiability crisis that MMD ignores and avoiding the collapse mode that Random Pairing induces.

H.5 Extended Results: Impact of Optimal Transport on Downstream Tasks

While the primary motivation for integrating OT is to prevent mean collapse and preserve population structure during training, its ultimate value lies in improving performance on downstream biological tasks. We conducted an ablation study comparing the standard OT-based pairing against a Random Pairing baseline to isolate the impact of this preprocessing step on the model’s ability to predict non-additive genetic interactions.

We evaluated both models on the “Genetic Interaction” benchmark (Precision @ 10). As shown in Table S8, the inclusion of OT leads to a marked improvement in identifying complex interaction types, specifically Redundancy, Epistasis, and Suppression.

Table S8: Ablation Study: Impact of Optimal Transport on Genetic Interaction Prediction (Precision @ 10). The model trained with OT preprocessing consistently outperforms the Random Pairing baseline in detecting complex interaction types (Redundancy, Epistasis, Suppression), demonstrating that OT helps preserve the fine-grained causal structure required for these tasks.

Interaction Type	Random Pairing	Optimal Transport
Neomorphism	0.4	0.4
Redundancy	0.7	0.8
Epistasis	0.8	0.9
Synergy	0.5	0.4
Suppression	0.2	0.3

The results indicate that while Random Pairing can achieve competitive performance on simpler metrics (like Neomorphism), it falls short in capturing the subtle dependencies required to identify Redundancy and Epistasis. The OT pairing, by matching cells based on their distributional similarity, likely preserves the underlying biological signal that defines these interactions, preventing them from being washed out by the noise of random assignment. This validates the hypothesis that OT is not just a theoretical necessity for causal identifiability but a practical enhancer of model utility for complex biological discovery.

H.6 Extended Results: Ablation Study on β -VAE and DAGMA Regularization

The primary goal of this ablation study is to rigorously assess the sensitivity of the RAPTORGraph model to its key regularization parameters, specifically the β parameter of the β -VAE (β) and the weights of the DAGMA loss (μ and λ_1). Our core hypotheses for this study were: 1. The criticality of β for preventing mode collapse, where we investigated if β is the most crucial hyperparameter for maintaining a regularized and informative latent space, looking for evidence of KL divergence approaching zero and its detrimental effects on learning meaningful representations. 2. The hierarchy of hyperparameter importance, testing the assertion that β is of primary importance, followed by the DAGMA weights, in terms of their impact on downstream performance.

The study was conducted using the RAPTORGraph model on the Norman et al. 2019 dataset. For the hyperparameter sweep, β was swept across 0.0, 0.1, 0.4, 0.8, 1.2, 1.6, 2.0, 2.5, 3.0 to observe its effect on KL divergence and to test for mode collapse. Additionally, μ and λ_1 were varied around default values to analyze their influence on the causal graph structure, specifically $\mu \in \{0.1, 0.27, 0.5\}$ and $\lambda_1 \in \{0.02, 0.05, 0.1\}$. The evaluation metrics included: a) KL Divergence as the primary metric to monitor for signs of mode collapse; b) Prediction Variance (Pred. Var.), representing the variance of the reconstructed gene expression for *control* cells, used to detect mean collapse; and c) MMD to evaluate the quality of predictions for *intervened or perturbed* cells. Notably, MSE is not explicitly used for mode collapse detection in this context, as variance directly assesses the diversity of generated outputs, which MSE alone might obscure.

To better visualize the dominant effect of β on the latent space, Table S9 summarizes all experiments. The goal is to demonstrate that β is the primary factor driving the KL divergence towards zero regardless of the DAGMA hyperparameter settings, indicating posterior collapse.

The consolidated table makes the trend clear. For a given β value (e.g., 0.4), the KL Divergence remains consistently low (≈ 0.004) across all variations of μ and λ_1 . However, changing β from 0.1 to 0.4, and then to 0.8 and higher, causes the KL Divergence to drop by orders of magnitude. This demonstrates that β is the determining factor for the degree of regularization and subsequent posterior collapse. The DAGMA weights have a negligible effect on the KL Divergence itself, reinforcing the conclusion that an appropriate β must be selected first before fine-tuning the other parameters.

Our argument is to choose $\beta = 0.4$ (approx. 0.447 in our final model) to balance the trade-off between distribution matching (MMD) and preserving biological heterogeneity. The KL Divergence at this range ($\approx 1 \times 10^{-3}$) is chosen to be similar to baselines like dVAE and SENA, ensuring the model captures meaningful biological variability rather than regressing to the population mean. While $\beta = 0.1$ preserves more variance, it suffers from poor stability and high MMD. Conversely, $\beta = 0.8$ achieves competitive MMD but suffers from an order-of-magnitude drop in prediction variance (1×10^{-4} vs 1×10^{-3}), indicating severe over-smoothing. We select $\beta = 0.4$ as the

Table S9: Impact of β -VAE and DAGMA Regularization on Model Performance. **Pred. Var.** denotes the variance of the predicted cell states (or variance of predicted control cells), where a lower value may indicate mean collapse. Lower MMD indicates better distribution matching. The bold row indicates the selected best configuration.

η	μ	λ_1	KL Divergence	Pred. Var.	MMD
0.0	0.27	0.052	0.0181	0.0181	1.929
0.1	0.1	0.02	0.059	0.0059	0.268
0.1	0.1	0.05	0.058	0.0067	0.257
0.1	0.1	0.1	0.054	0.0073	0.161
0.1	0.27	0.02	0.061	0.0051	0.401
0.1	0.27	0.05	0.060	0.0062	0.318
0.1	0.27	0.1	0.0615	0.0068	0.209
0.1	0.5	0.02	0.0609	0.0048	0.510
0.1	0.5	0.05	0.0604	0.0054	0.450
0.1	0.5	0.1	0.0572	0.0058	0.319
0.4	0.1	0.02	0.0040	0.0010	0.120
0.4	0.1	0.05	0.0039	0.0010	0.117
0.4	0.1	0.1	0.0040	0.0009	0.117
0.4	0.27	0.02	0.0040	0.0009	0.133
0.4	0.27	0.05	0.0039	0.0009	0.127
0.4	0.27	0.1	0.0040	0.0008	0.121
0.4	0.5	0.02	0.0040	0.0007	0.203
0.4	0.5	0.05	0.0040	0.0007	0.196
0.4	0.5	0.1	0.0040	0.0007	0.209
0.8	0.1	0.02	0.0009	0.0002	0.136
0.8	0.1	0.05	0.0009	0.0002	0.132
0.8	0.1	0.1	0.0009	0.0002	0.127
0.8	0.27	0.02	0.0009	0.0002	0.147
0.8	0.27	0.05	0.0009	0.0002	0.146
0.8	0.27	0.1	0.0009	0.0002	0.143
0.8	0.5	0.02	0.0008	0.0001	0.215
0.8	0.5	0.05	0.0009	0.0001	0.227
0.8	0.5	0.1	0.0009	0.0001	0.206
1.2	0.27	0.052	0.0003	0.00005	0.137
1.6	0.27	0.052	0.0002	0.00002	0.144
2.0	0.27	0.052	0.0001	0.00001	0.155
2.5	0.27	0.052	0.0001	0.000007	0.171
3.0	0.27	0.052	0.0000	0.000004	0.167

1225 optimal operating point, minimizing MMD while retaining significantly higher variance than the fully
1226 collapsed regimes. The bolded row in Table S9 represents this chosen configuration.

1227 **I Large Language Model (LLM) Usage Disclosure**

1228 We used Google Gemini 2.5 Pro to assist with proofreading and summarizing related work. All
1229 content was verified by the authors.